DAY 3 · WEDNESDAY 18 NOVEMBER 2026

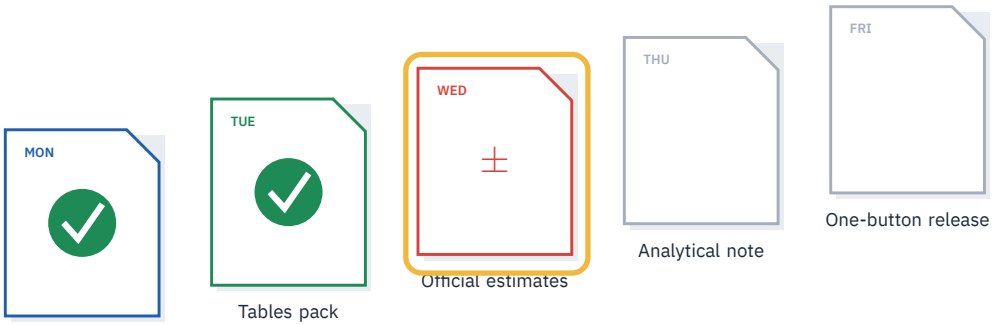
# From the sample to Seychelles

## Complex survey data and design-based estimation



In-country Stata Training Programme
National Bureau of Statistics, Seychelles
Assoc. Prof. Endri Raço, PhD

# Five days, five products



YOUR TURN · TASK 1

# Today’s files

**DO**

Type:

```
dir "C:/NBS_Stata/day3"
```

**YOU SHOULD SEE**

The day’s files: two labs, the practice set, the daily check and the `catchup` folder.

**IF NOT**

Nothing listed? Copy the `day3` folder from the USB stick into `C:\NBS_Stata`.

Yesterday's tables described the people we interviewed. Today's estimates describe Seychelles.

WEDNESDAY MORNING AT NBS

# The Director needs the Q1 2026 labour release

**QUESTION 1**

How many people does the survey stand for, and how many of them are employed?

**QUESTION 2**

What is the unemployment rate, nationally and in each region, with its margin of error?

**QUESTION 3**

Which regional figures are precise enough to publish?

By the end of today one do-file answers all three and writes the estimates table to Excel.

# What you will be able to do today

**AFTER THIS PART YOU CAN**

- explain what a survey weight is and check NBS's weights against the design
- declare NBS's sample design to Stata with `svyset`
- estimate means, totals, shares and rates with their standard errors
- produce regional and youth rates the correct way, with `subpop()`
- judge precision (CV, design effect) and export an estimates table to Excel



1

DAY 3 · PART 1

# Every respondent stands for others

THE PROBLEM

# Is 52.5% women in the file the share of women in Seychelles?

**WHAT WE HAVE**

5,888 persons interviewed: 2,796 men and 3,092 women (yesterday's `tabulate`).

**WHAT WE NEED**

The share among all people in the survey's scope, not among the people we happened to visit.

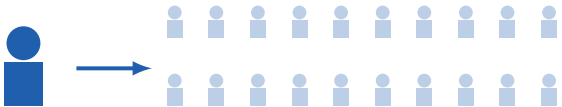
**WHY IT CAN DIFFER**

Some districts were sampled more heavily than others. Their people are over-represented in the file.

The weight corrects for that.

Every respondent stands for other people. The weight says how many.

# One interview, many people



one respondent

weight 20: she stands for 20 people

# The final weight on every person:

`summarize` `wgt_final`



YOUR TURN · TASK 2

# Look at the final weight

**DO**

Add to `my_day3.do` and run:

```
use "data/lfs_teach.dta", clear
describe wgt_base wgt_adj wgt_final
summarize wgt_final
```

**YOU SHOULD SEE**

Three float variables with labels; 5,888 weights between about 5 and about 303.

**IF NOT**

`file not found`? Run Day 1's catch-up: `do day1/catchup/catchup_D1_S9.do`.

# Read the weights

```
. summarize wgt_final

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
   wgt_final |      5,888    17.21534    12.39912       5.28    302.722  (1)
```

(1) on average one person stands for 17; the largest weight is 303

# Why the weights differ

**SELECTION**

Each district is sampled separately. A small district may get as many PSUs as a large one: its people get small weights; a large district's people get large ones.

**FIELDWORK**

Some districts were interviewed in one month of the quarter, others in two. Fewer interviews, larger weights.

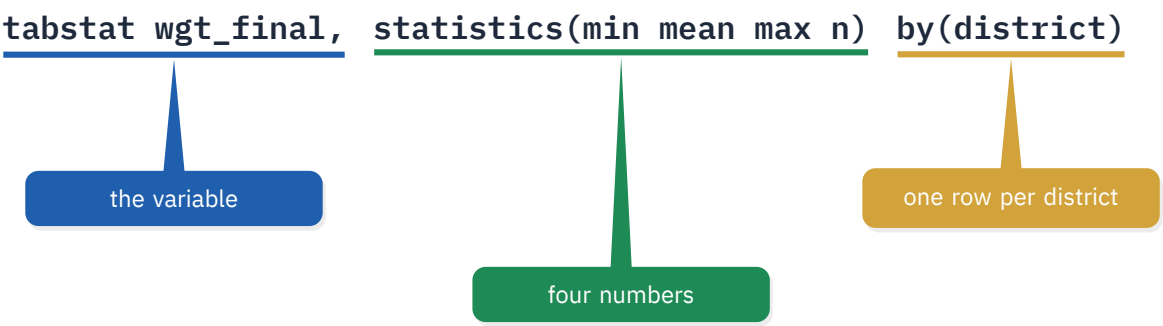
**ADJUSTMENT**

NBS then multiplies by a district adjustment so the weights add up to the population. That is `wgt_adj`.

**SOURCE**

UN, *Designing Household Survey Samples* (2005), ch. 5–6.

# The weight, district by district:



YOUR TURN · TASK 3

# Weights by district

**DO**

Type:

```
tabstat wgt_final, statistics(min mean max n)
by(district)
```

**YOU SHOULD SEE**

26 districts plus a Total row: smallest, mean and largest weight, and the number of persons.

**IF NOT**

`variable district not found`? You are not on `lfs_teach.dta`: run Task 2 again.

**EXTRA** Find the district whose persons have the smallest mean weight.

# Read the district weights

| district | Min | Mean | Max | N | |
|---|---|---|---|---|---|
| Les Mamelles | 5.472445 | 8.169747 | 11.9612 | 288 | 1 |
| Anse Royale | 29.69934 | 37.31533 | 45.89898 | 133 | 2 |
| Anse Boileau | 17.69155 | 39.44964 | 302.722 | 119 | 3 |
| Total | 5.28 | 17.21534 | 302.722 | 5888 | 4 |

1. Les Mamelles: each person stands for 8, on average
2. Anse Royale: each stands for 37
3. Anse Boileau: mean 39, but one weight of 303
4. all 5,888: from 5.28 to 302.7

QUICK CHECK

## A person in Les Mamelles (mean weight 8) and a person in Anse Royale (mean weight 37). Who stands for more people in Seychelles?

A. the person in Les Mamelles

B. the person in Anse Royale

C. the same: one interview is one interview

ANSWER

## A person in Les Mamelles (mean weight 8) and a person in Anse Royale (mean weight 37). Who stands for more people in Seychelles?

A the person in Les Mamelles

**B the person in Anse Royale**

C the same: one interview is one interview

An Anse Royale interview stands for about four times as many people. Unweighted tables give both the same voice.

# How many people do the weights add up to?

```
summarize wgt_final
```

stores the results

```
display r(sum)
```

the sum of all 5,888 weights (Day 2: r())

YOUR TURN · TASK 4

# Add up the weights

**DO**

Type:

```
summarize wgt_final
display r(sum)
```

```
display %12.0fc r(sum)
```

**YOU SHOULD SEE**

One number with decimals, then the same number with a thousands comma.

**IF NOT**

`r(sum)` shows a dot? Run `summarize` directly before `display`.

**EXTRA** What does `%12.0fc` change?

# The population behind the sample

```
. display r(sum)
101363.92 (1)

. display %12.0fc r(sum)
     101,364 (2)
```

1. the weights add up to 101,363.92
2. rounded, with a comma: 101,364

**5,888 interviews stand for 101,364 people in the survey's scope.**

# Same file, two answers

**UNWEIGHTED: THE INTERVIEWS**

```
. tabulate ilo_sex

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
   1 - Male |      2,796       47.49       47.49
 2 - Female |      3,092       52.51      100.00
------------+-----------------------------------
      Total |      5,888      100.00
```

**WEIGHTED: SEYCHELLES**

```
. tabulate ilo_sex [iweight = wgt_final]

        Sex |      Freq.     Percent        Cum.
------------+-----------------------------------
   1 - Male | 48,242.259       47.59       47.59
 2 - Female | 53,121.662       52.41      100.00
------------+-----------------------------------
      Total | 101,363.92      100.00
```

1 52.51% of interviews are women. 2 52.41% of the population is. Small here; on other variables the gap is larger.

# A quick weighted count:

```
tabulate region [iweight = wgt_final]
```

yesterday's table

each row counts as its weight

YOUR TURN · TASK 5

# People by region

**DO**

Type:

```
tabulate region [iweight = wgt_final]
```

**YOU SHOULD SEE**

Seven regions with counts in thousands, adding to 101,363.92.

**IF NOT**

`pweights not allowed`? You typed `pweight`: use `iweight` here.

**EXTRA** Which region holds a quarter of the population?

# Read the weighted table

| Region | Freq. | Percent | Cum. | |
|---|---|---|---|---|
| 1- North | 18,146.795 | 17.90 | 17.90 | |
| 2- Central 1 | 12,875.658 | 12.70 | 30.61 | |
| 3- Central 2 | 12,228.789 | 12.06 | 42.67 | |
| 4- EastSouth | 25,820.712 | 25.47 | 68.14 | 1 |
| 5- West | 15,443.634 | 15.24 | 83.38 | |
| 6- Praslin & LaDigue | 11,139.212 | 10.99 | 94.37 | |
| 7- Perseverance | 5,709.1193 | 5.63 | 100.00 | 2 |
| Total | 101,363.92 | 100.00 | | 3 |

1. EastSouth: 25,821 people, a quarter of the total
2. Perseverance: 5,709
3. the total is the sum of the weights

COMMON ERROR

# A survey weight in a plain table

```
. tabulate ilo_sex [pweight = wgt_final]
pweights not allowed
```

**WHY**

In `tabulate`, the probability weight `pweight` is not allowed. Stata reserves it for commands that compute standard errors.

**FIX**

For a quick count use `[iweight = wgt_final]`. For a published figure use `svy: tabulate` (Part 6).

COMMON ERROR

# One letter missing

```
. summarize wgt_finl
variable wgt_finl not found
```

**WHY**

There is no variable `wgt_finl`. Stata does not guess.

**FIX**

Copy the name from the Variables window, or type `wgt_f` and press Tab.

# Weights move the mean too

**PLAIN MEAN**

```
. mean ilo_age

Mean estimation                         Number of obs = 5,888

------------------------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |   36.69497   .2861558        36.134    37.25594
------------------------------------------------------------------------------
```

1

**WEIGHTED MEAN**

```
. mean ilo_age [pweight = wgt_final]

Mean estimation                         Number of obs = 5,888

------------------------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |   37.05894   .3389491      36.39448    37.72341
------------------------------------------------------------------------------
```

2

1 the interviews average 36.7 years. 2 the population averages 37.1. The standard error changes too, but it is not yet right: the design is missing.

WARNING

# A weight alone fixes the estimate, not the margin of error.

The standard error also depends on how the sample was drawn: districts as strata, PSUs as clusters.

**Weights + design = `svyset` (Part 3).**

ON YOUR OWN · A

# Which region carries the most weight?

**THE QUESTION**

For each region, show the smallest, mean and largest weight and their sum. Which region stands for the most people, and how many?

**HINT**

`tabstat` with `statistics(min mean max sum)`, `by(region)`; add `format(%9.0fc)`.

ON YOUR OWN · A · ANSWER

```
tabstat wgt_final, statistics(min mean max sum) ///
by(region) format(%9.0fc)
```

| region | Min | Mean | Max | Sum |
|---|---|---|---|---|
| 1- North | 9 | 19 | 37 | 18,147 |
| 2- Central 1 | 5 | 15 | 33 | 12,876 |
| 3- Central 2 | 5 | 15 | 27 | 12,229 |
| 4- EastSouth | 7 | 18 | 46 | 25,821 |
| 5- West | 5 | 17 | 303 | 15,444 |
| 6- Praslin & LaD | 12 | 22 | 84 | 11,139 |
| 7- Perseverance | 9 | 16 | 37 | 5,709 |
| Total | 5 | 17 | 303 | 101,364 |

1. EastSouth: 25,821 people
2. West holds the 303 weight

QUICK CHECK

## The 5,888 weights add up to 101,364. What is that number?

A. the number of interviews

B. the population the survey stands for

C. the number of households

ANSWER

## The 5,888 weights add up to 101,364. What is that number?

A the number of interviews

**B the population the survey stands for**

C the number of households

Each weight is a number of people; their sum is the household population in scope. Households are 1,858.

QUICK CHECK

## Which command gives a figure you can publish, with its standard error?

A. tabulate ilo_sex [iweight = wgt_final]

B. mean ilo_age [pweight = wgt_final]

C. svy: mean ilo_age, after svyset

ANSWER

## Which command gives a figure you can publish, with its standard error?

A tabulate ilo_sex [iweight = wgt_final]

B mean ilo_age [pweight = wgt_final]

**C** svy: mean ilo_age, after svyset

Only `svy:` uses the weights and the design together. Parts 3 and 4.

# Part 1 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `summarize wgt_final` | smallest, mean, largest weight |
| `display r(sum)` | the population the weights stand for |
| `tabstat w, statistics(min mean max) by(g)` | weights by group |
| `tabulate v [iweight = w]` | a quick weighted count, no SE |
| `mean v [pweight = w]` | weighted mean, SE still wrong |
| `svyset ... (Part 3)` | weights + design: the right SE |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 1**

CORE P1 Weighted count of men and women (`iweight`). Type the weighted number of women.

CORE P2 Sum of the weights by region with `tabstat`. Type the sum for Perseverance.

CORE P3 Weighted share of women, one decimal.

EXTENSION P4 Which district has the largest single weight? Type it.

CHALLENGE P5 List the record(s) with a weight above 100: district, PSU, household.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# You can explain what a weight is, and why it changes the answer.

lost? open `catchup_D3_S1.do`

2

DAY 3 · PART 2

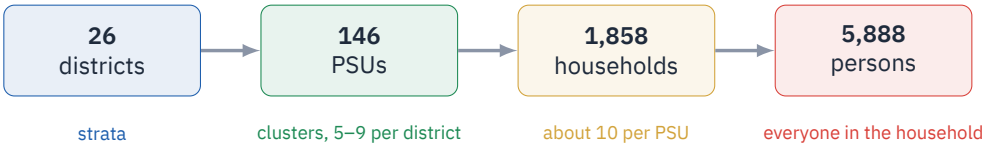
# How the sample was drawn

# Part 2: the design behind the weights

**AFTER THIS PART YOU CAN**

- name the strata, the PSUs and the households of the LFS
- count PSUs per district with `egen tag()`
- prove that `wgt_final` = `wgt_base` × `wgt_adj`
- check that a value is constant within a group with `[1]`

# From districts to persons



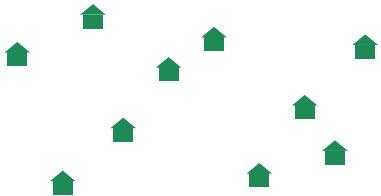
# Districts are the strata: every district is sampled on its own.

**PSUs are the clusters: households are drawn inside them, not across the whole island.**

# Ten neighbours tell you less than ten strangers



ten households in one PSU:
similar jobs, similar income



ten households across Mahé:
more variety, more information

Clustering makes the standard error larger than in a simple random sample. The ratio of the two variances is the **design effect** (Part 4).

# Count each PSU once:

```
egen first_psu = tag(district psu)
```

a new 0/1 variable

1 on the first row of each PSU, 0 on the others

# Then count the PSUs of each district

```
egen first_psu = tag(district psu)
bysort district: egen n_psu = total(first_psu)
tabulate n_psu if first_psu
```

- the first row of each PSU is tagged 1
- adding the tags in a district counts its PSUs (egen total, Day 1)
- `if first_psu`: one row per PSU, so the table counts PSUs, not persons

YOUR TURN · TASK 6

# PSUs per district

**DO**

Type the three lines:

```
egen first_psu = tag(district psu)
bysort district: egen n_psu = total(first_psu)
tabulate n_psu if first_psu
```

**YOU SHOULD SEE**

A small table: how many PSUs have 5, 6, 7 or 9 PSUs in their district; Total 146.

**IF NOT**

`variable first_psu already defined`? You ran the line twice: `drop first_psu n_psu` and run again.

**EXTRA** How many districts have exactly 7 PSUs?

# Read the PSU table

| n_psu | Freq. | Percent | Cum. | |
|---|---|---|---|---|
| 5 | 90 | 61.64 | 61.64 | 1 |
| 6 | 12 | 8.22 | 69.86 | |
| 7 | 35 | 23.97 | 93.84 | |
| 9 | 9 | 6.16 | 100.00 | 2 |
| Total | 146 | 100.00 | | 3 |

1. 90 PSUs are in districts with 5 PSUs: 18 districts
2. one district has 9 PSUs
3. 146 PSUs in all; the smallest district has 5

# PSUs, district by district



| district | psus |
|---|---|
| English River | 5 |
| Mont Buxton | 5 |
| Saint Louis | 5 |
| Bel Air | 7 |
| Mont Fleuri | 5 |
| Plaisance | 7 |
| Roche Caiman | 5 |
| Les Mamelles | 5 |
| Ile Perseverance | 7 |
| Cascade | 5 |
| Pointe Larue | 6 |
| Anse Aux Pins | 7 |
| Anse Royale | 5 |
| Takamaka | 5 |
| Au Cap | 6 |
| Baie Lazare | 5 |
| Anse Boileau | 5 |
| Grande Anse Mahe | 7 |
| Port Glaud | 5 |
| Belombre | 5 |
| Beau Vallon | 5 |
| Glacis | 5 |
| Anse Etoile | 5 |
| Grande Anse Praslin | 5 |
| Baie Ste Anne | 9 |
| La Digue and Inner islands | 5 |

1

1. Baie Ste Anne (Praslin) has 9 PSUs; every other district 5 to 7

# Two PSUs per district is the minimum for a standard error.

With one PSU, Stata cannot measure how much PSUs differ inside the district.

**Here every district has at least 5.**

# Three weights, one product




from the selection probabilities, constant in a PSU

NBS's district adjustment, constant in a district

the weight for every estimate

YOUR TURN · TASK 7

# Rebuild the final weight

**DO**

Type:

```
generate check_w = wgt_base * wgt_adj
assert abs(check_w - wgt_final) < 1e-4
summarize check_w wgt_final
```

**YOU SHOULD SEE**

The assert is silent. Both rows of the summary are identical.

**IF NOT**

`assertion is false`? Check the names: base times adjustment.

**EXTRA** Why `abs(...) < 1e-4` rather than `==`?

# The proof, in two lines

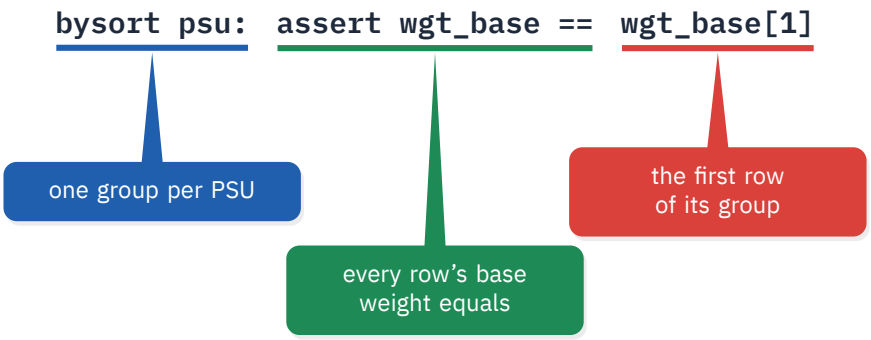
```
. generate check_w = wgt_base * wgt_adj

. assert abs(check_w - wgt_final) < 1e-4
. summarize check_w wgt_final

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
     check_w |      5,888    17.21534    12.39912       5.28    302.722
   wgt_final |      5,888    17.21534    12.39912       5.28    302.722
```

1. silent: true for all 5,888 rows
2. same mean, same smallest, same largest

# Is the base weight the same in the whole PSU?



YOUR TURN · TASK 8

# Base weight constant in each PSU

**DO**

Type:

```
bysort psu: assert wgt_base == wgt_base[1]
```

**YOU SHOULD SEE**

Nothing: silence means every PSU has a single base weight.

**IF NOT**

`not sorted`? You typed `by`, not `bysort`.

# What [1] looks at

```
. list psu unique_ind wgt_base in 1/6

     +--------------------------------+
     |      psu   uniqu~nd   wgt_base |
     |--------------------------------|
  1. | 11211121    1121101        4.2 |
  2. | 11211121    1121102        4.2 |
  3. | 11211121    1121103        4.2 |
  4. | 11211121     112111        4.2 |
  5. | 11211121     112112        4.2 |
     |--------------------------------|
  6. | 11211121     112113        4.2 |
     +--------------------------------+
```

1. row 1 of PSU 11211121: base weight 4.2
2. every other row of the PSU must equal it

COMMON ERROR

# by without sorting first

```
. by psu: assert wgt_base == wgt_base[1]
not sorted
```

**WHY**

by needs the data already sorted by the group. After Task 5 the file is sorted by district, not by PSU.

**FIX**

Use `bysort`: it sorts, then runs.

YOUR TURN · TASK 9

# Adjustment constant in each district

**DO**

Type:

```
bysort district: assert wgt_adj == wgt_adj[1]
```

**YOU SHOULD SEE**

Silence.

**IF NOT**

`assertion is false`? Tell the trainer: NBS's file would have changed.

**EXTRA** Show the adjustment of each district with `tabstat ..., statistics(min max) by(district)`.

# One adjustment per district

| district | Min | Max | |
|---|---|---|---|
| English River | 3.45 | 3.45 | 1 |
| Mont Buxton | 2.76 | 2.76 | |
| Saint Louis | 2.999 | 2.999 | |
| Bel Air | 1.65 | 1.65 | |
| Total | 1.65 | 4.5 | 2 |

1. min = max in every district: one factor each
2. from 1.65 (Bel Air) to 4.5 (Anse Etoile)

# The largest weight in the file

```
. list district psu unique_hhd wgt_base wgt_adj wgt_final if wgt_final > 100

     +-------------------------------------------------------------------------+
     |     district        psu   uniqu~hd   wgt_base   wgt_adj   wgt_fi~l |
     |-------------------------------------------------------------------------|
3671. | Anse Boileau   33263328     332835         98     3.089    302.722 |
3672. | Anse Boileau   33263328     332835         98     3.089    302.722 |
3673. | Anse Boileau   33263328     332835         98     3.089    302.722 |
3674. | Anse Boileau   33263328     332835         98     3.089    302.722 |
3675. | Anse Boileau   33263328     332835         98     3.089    302.722 |
     +-------------------------------------------------------------------------+
```

1. one household in Anse Boileau, 5 persons
2. base weight 98 times the district factor 3.089 = 302.7

WARNING

# For estimates, use `wgt_final`.

`wgt_base` misses NBS's adjustment: it adds up to a different population (Part 7). `ilo_wgt`, dropped on Day 1, was 1 for everyone.

**One weight in every svyset line: `wgt_final`.**

For colleagues at the SADC sampling workshop, Johannesburg, March 2026: **the same design logic, now on NBS's own survey.**

ON YOUR OWN · B

# How many households does a PSU hold?

THE QUESTION

Count the households in each PSU and describe the distribution: smallest, median, largest.

HINT

`egen ... = tag(psu unique_hhd)`, then `bysort psu: egen ... = total(...)`, then `summarize ... if first_psu, detail`.

ON YOUR OWN · B · ANSWER

```
egen first_hh = tag(psu unique_hhd)
bysort psu: egen n_hh = total(first_hh)
summarize n_hh if first_psu, detail
```

```
. summarize n_hh if first_psu, detail

                            n_hh
-------------------------------------------------------------
      Percentiles      Smallest
 1%             4             1
 5%             7             4
10%             8             6       Obs                 146
25%            10             6       Sum of wgt.         146

50%            10                     Mean           12.72603   (1)
                        Largest       Std. dev.      5.117274
75%            18            23
90%            20            23       Variance       26.18649
95%            21            23       Skewness       .6562674
99%            23            28       Kurtosis       2.351789   (2)
```

1. median 10 households per PSU, the design target
2. from 1 to 28

QUICK CHECK

## In `bysort psu: assert wgt_base == wgt_base[1]`, what is `wgt_base[1]`?

A. the base weight of the first row of the file

B. the base weight of the first row of the PSU

C. the smallest base weight

ANSWER

## In `bysort psu: assert wgt_base == wgt_base[1]`, what is `wgt_base[1]`?

A. the base weight of the first row of the file

**B. the base weight of the first row of the PSU**

C. the smallest base weight

Under `by`, row numbers restart in each group: `[1]` is the first row of the current PSU.

QUICK CHECK

## After `egen first_psu = tag(district psu)`, how many rows have first_psu = 1?

A. 26

B. 146

C. 5,888

ANSWER

## After `egen first_psu = tag(district psu)`, how many rows have first_psu = 1?

A 26

**B 146**

C 5,888

One per PSU: 146 (log D02, Total).

# Part 2 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `egen t = tag(a b)` | 1 on the first row of each a-b combination |
| `bysort g: egen n = total(t)` | count tagged rows per group |
| `tabulate n if t` | one row per group in the table |
| `assert abs(x - y) < 1e-4` | equal, allowing for rounding |
| `bysort g: assert x == x[1]` | x is constant within g |
| `v[1] v[_N]` | first and last row of the group |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 2**

| | | |
|---|---|---|
| CORE | P1 | How many PSUs are in the file? Use `egen tag()` and `count`. |
| CORE | P2 | Which district has the most PSUs? Type how many. |
| CORE | P3 | Prove `wgt_final` = `wgt_base` × `wgt_adj` with one assert. |
| EXTENSION | P4 | Median number of households per PSU. |
| CHALLENGE | P5 | Prove with one line that every household lies in one PSU only. |

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# You have checked NBS's weights against the design, with Stata doing the proof.

lost? open **catchup_D3_S2.do**



3

DAY 3 · PART 3

# Declaring the design

COMMON ERROR

# The survey prefix, before the design

```
. svy: mean ilo_age
data not set up for svy, use svyset
```

**WHY**

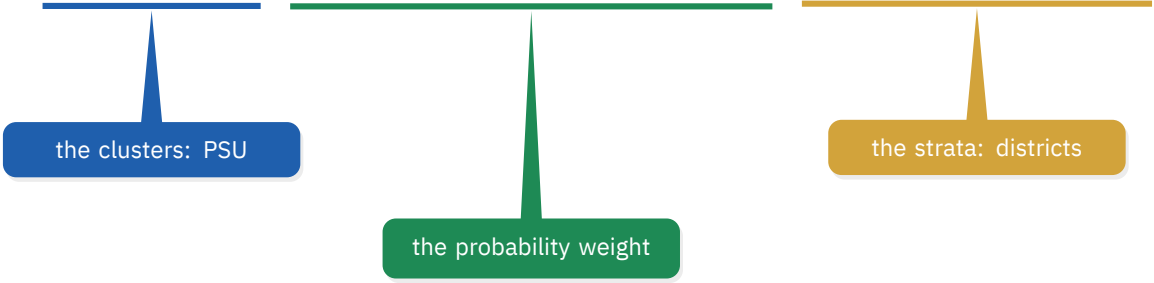
Stata does not know which variables hold the PSU, the strata and the weight. Without them it cannot compute a survey estimate.

**FIX**

Declare the design once with `svyset`. Every `svy:` command then uses it.

# One line declares the design:

```
svyset psu [pweight = wgt_final], strata(district)
```



# `pweight`: the probability weight.

The only weight type for survey estimates: the inverse of each person's chance of selection, adjusted.

**Short form: `pw`.**

# Design first, options after the comma.

PSU and weight before the comma; `strata()` and other options after it.

**Forget the comma and Stata reads strata() as a variable name.**

YOUR TURN · TASK 10

# Declare the design

**DO**

Add to `my_day3.do`:

```
svyset psu [pweight = wgt_final], strata(district)
```

**YOU SHOULD SEE**

Seven lines: the weight, VCE linearized, the strata variable, the sampling unit.

**IF NOT**

Red text? Read it on the next slides: two common errors.

# Read the declaration

```
Sampling weights: wgt_final (1)
             VCE: linearized (2)
      Subpop. df: off
     Single unit: missing
        Strata 1: district (3)
 Sampling unit 1: psu (4)
           FPC 1: <zero> (5)
```

1. the weight
2. linearised standard errors (Taylor series)
3. districts are the strata
4. PSUs are the clusters
5. no finite population correction

COMMON ERROR

# A misspelled strata variable

```
. svyset psu [pweight = wgt_final], strata(distrct)
variable distrct not found
(error in option strata())
```

**WHY**

Stata checks every name in `svyset`. `distrct` does not exist.

**FIX**

Fix the name and run the line again. The old declaration stays until a new one succeeds.

COMMON ERROR

# svyset without the comma

```
. svyset psu [pweight = wgt_final] strata(district)
invalid 'strata'
```

**WHY**

Without the comma, Stata reads `strata(district)` as part of the design and refuses it.

**FIX**

Put the comma after the weight: `[pweight = wgt_final], strata(district)`.

# Check the design Stata now holds:

**`svydescribe`**

strata, PSUs and observations per PSU

YOUR TURN · TASK 11

# Describe the design

**DO**

Type:

```
svydescribe
```

**YOU SHOULD SEE**

One row per district (stratum code), the number of PSUs and observations, and a total line: 26, 146, 5,888.

**IF NOT**

`data not set up for svy`? Run Task 9 first.

**EXTRA** Find the stratum with the most observations per PSU.

# Read svydescribe

| | | | Number of obs per unit | | | |
|---|---|---|---|---|---|---|
| Stratum | # units | # obs | Min | Mean | Max | 1 |
| 11 | 5 | 304 | 50 | 60.8 | 71 | 2 |
| 12 | 5 | 158 | 22 | 31.6 | 46 | |
| 13 | 5 | 153 | 18 | 30.6 | 46 | |
| 26 | 146 | 5,888 | 5 | 40.3 | 99 | 3 |

1. one row per stratum (district code)
2. district 11: 5 PSUs, 304 persons
3. 26 strata, 146 PSUs, 5,888 persons; 5 to 99 per PSU

# svydescribe for one variable

| Stratum | Number of units included | Number of units omitted | Number of obs with complete data | Number of obs with missing data | # obs per included unit Min | # obs per included unit Mean | # obs per included unit Max | |
|---|---|---|---|---|---|---|---|---|
| 11 | 5 | 0 | 161 | 143 | 28 | 32.2 | 39 | 1 |
| 12 | 5 | 0 | 80 | 78 | 10 | 16.0 | 22 | |
| 13 | 5 | 0 | 78 | 75 | 8 | 15.6 | 25 | |
| 26 | 146 | 0 | 2,999 | 2,889 | 4 | 20.5 | 48 | 2 |
| | | | | 5,888 | | | | |

1. district 11: 161 persons with usual hours, 143 without
2. 2,999 answers in all; every PSU keeps at least 4

YOUR TURN · TASK 12

# Save the declared file

**DO**

Type:

```
save "data/lfs_svy.dta", replace
use "data/lfs_svy.dta", clear
svyset
```

**YOU SHOULD SEE**

The file is saved; after opening it again, `svyset` alone shows the same design.

**IF NOT**

`file ... could not be opened`? Close the file if it is open in another program.

# The design travels with the file

```
. svyset 1

Sampling weights: wgt_final
             VCE: linearized
      Subpop. df: off
     Single unit: missing
        Strata 1: district 2
 Sampling unit 1: psu
           FPC 1: <zero>
```

1. `svyset` with nothing after it: show the stored declaration
2. same strata, PSU and weight as before saving

COMMON ERROR

# A weight inside a svy command

```
. svy: mean ilo_age [pweight = wgt_final]
weights not allowed with the svy prefix
    The svy prefix assumes survey weights were already specified using svyset.
```

**WHY**

The weight is already part of the design. Giving it again is refused.

**FIX**

Write `svy: mean ilo_age` with no weight.

# Finite population correction (recognise it)

When a large share of the PSUs in a stratum is sampled, the variance is smaller.

```
svyset psu [pweight = wgt_final], strata(district) fpc(N_psu)
```

`N_psu`: the number of PSUs in the district's frame. NBS's file has no such variable, so `FPC 1: <zero>`: no correction. The SE is slightly conservative.

# Replicate weights (recognise them)

Some offices ship variance information as replicate weights instead of PSUs and strata.

```
svyset [pweight = w], brrweight(rw1-rw80)
svyset [pweight = w], jkrweight(jk1-jk100)
```

BRR and jackknife. Same `svy:` commands afterwards. Not used here: NBS's file has one weight and the design variables.

WARNING

# A district with one PSU breaks the variance.

`svydescribe` would show it with `# units` = 1. In this file the smallest is 5.

**If it ever happens: merge the stratum with a similar one, with NBS's methodologist.**

QUICK CHECK

## You open `lfs_svy.dta` tomorrow. Must you run `svyset` again?

A. yes, every time Stata starts

B. no: the declaration was saved with the data

C. only if the file was renamed

ANSWER

## You open `lfs_svy.dta` tomorrow. Must you run `svyset` again?

A yes, every time Stata starts

**B** no: the declaration was saved with the data

C only if the file was renamed

`svyset` is stored in the .dta. Keep the line in the do-file anyway: it documents the design.

ON YOUR OWN · C

# Prove the design is stored

**THE QUESTION**

Declare the design, save the file as `data/lfs_svy.dta`, open it again, and show that Stata still knows the strata and the PSUs.

**HINT**

Three lines after `svyset`: `save`, `use`, `svyset` alone.

ON YOUR OWN · C · ANSWER

```
svyset psu [pweight = wgt_final], strata(district)
save "data/lfs_svy.dta", replace
use "data/lfs_svy.dta", clear
svyset
```

```
. svyset

Sampling weights: wgt_final
             VCE: linearized
      Subpop. df: off
     Single unit: missing
        Strata 1: district
 Sampling unit 1: psu
             FPC 1: <zero>
```

1 the stored design, read back from the file

QUICK CHECK

## Which line declares PSUs as clusters and districts as strata?

A. svyset district [pweight = wgt_final], strata(psu)

B. svyset psu [pweight = wgt_final], strata(district)

C. svyset [pweight = wgt_final], strata(district)

ANSWER

## Which line declares PSUs as clusters and districts as strata?

A svyset district [pweight = wgt_final], strata(psu)

**B** svyset psu [pweight = wgt_final], strata(district)

C svyset [pweight = wgt_final], strata(district)

Clusters before the weight, strata after the comma. C has no clusters: each person would be a PSU (log C07).

# Part 3 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `svyset psu [pweight = w], strata(s)` | declare the design |
| `svyset` | show the stored declaration |
| `svydescribe` | strata, PSUs, obs per PSU |
| `svydescribe v` | same, for rows where v is not missing |
| `... fpc(N)` | finite population correction (recognise) |
| `... brrweight() jkrweight()` | replicate weights (recognise) |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 3**

CORE P1 Declare the design and run `svydescribe`. Type the number of strata.

CORE P2 Type the smallest number of observations in a PSU.

CORE P3 Save the declared file and show the design after reopening it.

EXTENSION P4 `svydescribe ilo_job1_how_usual`: how many persons have complete data?

CHALLENGE P5 Declare the design without strata and run `svydescribe`: how many strata does Stata see?

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# Your file now carries NBS's sample design. Every estimate from here respects it.

lost? open **catchup_D3_S3.do**

# Lab 1 · Document the design

**THE SITUATION**

The methodology unit asks for a one-page note on the Q1 2026 LFS design, with every number produced by Stata.

**OPEN**

`lab_D3_1.do` in `C:\NBS_Stata\day3`.

**THE TASKS**

**CORE** Tasks 1 to 5: weights, PSUs per district, rebuilt weight, svyset, svydescribe

**EXTENSION** Task 6: PSUs per district to Excel

**CHALLENGE** Task 7: districts interviewed in one month only, and their weights

**CORRECT WHEN**

the whole file runs without red text and the five-line design note is written as comments.

# The lab file: lab_D3_1.do

```
* TASK 2 (Core)  The weights
*   Open lfs_teach.dta. Summarize wgt_final and display the sum of
*   the weights with a thousands comma.
* >>> YOUR CODE



* CHECK Task 2 (silent = correct)
quietly summarize wgt_final
assert abs(r(sum) - 101363.92) < 0.1
```

Type under `>>> YOUR CODE`, run the whole file, read the CHECK: silent means correct.

Green note when your last check is silent. Then the Challenge task.



4

DAY 3 · PART 4

# One prefix: svy:

THE DIRECTOR'S QUESTIONS, PART 1

# Four headline numbers for Q1 2026

**MEAN**

Average age of the population

**TOTAL**

How many people are employed

**SHARES**

Working-age population by labour force status

**RATE**

The unemployment rate

Each with its standard error and 95% confidence interval.

# Part 4: design-based estimates

**AFTER THIS PART YOU CAN**

- build four 0/1 flags that are never missing
- estimate a mean, a total, shares and a ratio with `svy:`
- read the standard error and the confidence interval
- read the design effect with `estat effects`

# Four flags, never missing

```
generate wap   = (ilo_age >= 15)
generate lf    = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
generate unemp = (ilo_lfs19 == 2)
generate emp   = (ilo_lfs19 == 1)
```

- `wap`: working age, 15 and over
- `lf`: in the labour force (employed or unemployed)
- `unemp`, `emp`: unemployed, employed
- a true/false test gives 1 or 0 on every row, also where `ilo_lfs19` is missing

YOUR TURN · TASK 13

# Build the four flags

**DO**

Add the four lines, then check one of them:

```
generate wap = (ilo_age >= 15)
generate lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)
generate unemp = (ilo_lfs19 == 2)
generate emp = (ilo_lfs19 == 1)

tabulate ilo_lfs19 lf, missing
```

**EXTRA** Run `misstable summarize wap lf unemp emp`: what does it list?

**YOU SHOULD SEE**

Employed and unemployed in column 1, everyone else in column 0, also the 1,261 under 15.

**IF NOT**

`variable lf already defined`? `drop wap lf unemp emp` and run the block again.

# Read the check

```
. tabulate ilo_lfs19 lf, missing

Labour Force Status - |          lf
            19th ICLS |         0          1 |     Total
----------------------+----------------------+----------
         1 - Employed |         0      3,012 |     3,012
       2 - Unemployed |         0        114 |       114
3 - Outside Labour Fo |     1,501          0 |     1,501
                    . |     1,261          0 |     1,261
----------------------+----------------------+----------
                Total |     2,762      3,126 |     5,888
```

1. employed and unemployed: `lf` = 1
2. the 1,261 without a status: `lf` = 0, not missing
3. 3,126 in the labour force

# The flags are never missing

```
. misstable summarize wap lf unemp emp ilo_lfs19

                                                              Obs<.
                                                +------------------------------
                |                               | Unique
     Variable   |     Obs=.     Obs>.     Obs<. | values        Min          Max
----------------+-----------------------------------+------------------------------
    ilo_lfs19   |     1,261               4,627 |      3          1            3
----------------------------------------------------------------------------------
```

1

1 only `ilo_lfs19` is listed: the four flags have no missing values

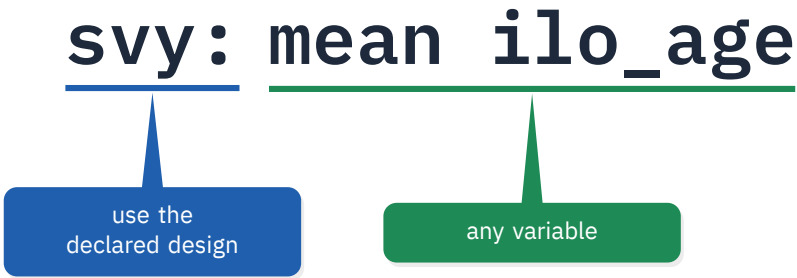
WHY FLAGS

# Rows with a missing value drop out of a `svy:` estimate.

Then the estimate describes a different population than you think. Flags with 0 and 1 everywhere keep every row; groups are chosen with `subpop()`.

**Build 0/1 flags first, estimate second.**

# A design-based mean:



YOUR TURN · TASK 14

# Mean age, design-based

**DO**

Type:

```
svy: mean ilo_age
```

**YOU SHOULD SEE**

A header with 26 strata and 146 PSUs, then one row: mean, linearised standard error, 95% interval.

**IF NOT**

`data not set up for svy`? Run the `svyset` line (Task 9) or open `lfs_svy.dta`.

# Read svy: mean

```
. svy: mean ilo_age
Number of strata =  26            Number of obs    =      5,888 (1)
Number of PSUs   = 146            Population size = 101,363.92 (2)
                                  Design df       =        120 (3)

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
     ilo_age |   37.05894   .4107522      36.24568    37.87221 (4)
--------------------------------------------------------------
```

1. the design used: 26 strata, 146 PSUs
2. the population: the sum of the weights
3. design degrees of freedom: PSUs minus strata
4. 37.06 years; SE 0.41; 95% CI 36.2 to 37.9

# Same variable, two commands

**mean ilo_age**

```
. mean ilo_age

Mean estimation                          Number of obs = 5,888

------------------------------------------------------------------------------
             |       Mean   Std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |   36.69497   .2861558         36.134    37.25594
------------------------------------------------------------------------------
```

1

**svy: mean ilo_age**

```
. svy: mean ilo_age

Number of strata =  26            Number of obs    =       5,888
Number of PSUs   = 146            Population size = 101,363.92
                                  Design df        =         120

------------------------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
     ilo_age |   37.05894   .4107522      36.24568    37.87221
------------------------------------------------------------------------------
```

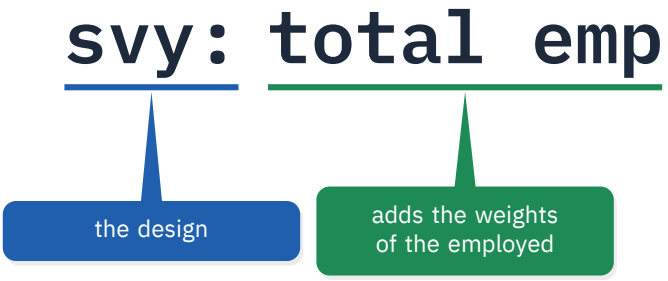
2

1 36.69, SE 0.29: the 5,888 interviews. honest margin.

2 37.06, SE 0.41: Seychelles, with the design's

The plain command describes the interviews. `svy:` describes Seychelles, with an honest margin of error.

# How many people are employed?



YOUR TURN · TASK 15

# Number of employed persons

**DO**

Type:

```
svy: total emp
```

**YOU SHOULD SEE**

About 52 thousand, with a standard error and an interval.

**IF NOT**

`variable emp not found`? Run Task 12.

**EXTRA** Round the interval to the nearest hundred: what would the release say?

# Read svy: total

```
. svy: total emp

Number of strata =  26            Number of obs   =      5,888
Number of PSUs   = 146            Population size = 101,363.92
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |      Total   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         emp |   52431.64   1715.623      49034.83    55828.46
--------------------------------------------------------------
```

1

1 52,432 employed; 95% CI 49,035 to 55,828

ON YOUR OWN · D

# How many people are unemployed?

**THE QUESTION**

Estimate the number of unemployed persons in Seychelles in Q1 2026, with its 95% interval.

**HINT**

Same command as the employed, on the other flag.

ON YOUR OWN · D · ANSWER

```
svy: total unemp
```

```
. svy: total unemp

Number of strata =  26            Number of obs    =      5,888
Number of PSUs   = 146            Population size = 101,363.92
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |      Total   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |   1792.994   180.1673      1436.276    2149.713
--------------------------------------------------------------
```

1

1. 1,793 unemployed; 95% CI 1,436 to 2,150

# Shares of the working-age population:

```
svy, subpop(wap): proportion ilo_lfs19
```

only persons 15+, keeping the whole design

one share per category

YOUR TURN · TASK 16

# Working-age population by status

**DO**

Type:

```
svy, subpop(wap): proportion ilo_lfs19
```

**YOU SHOULD SEE**

Three shares that add up to 1: employed, unemployed, outside the labour force.

**IF NOT**

Shares look wrong? Check `wap` with `tabulate wap`.

# Read svy: proportion

```
. svy, subpop(wap): proportion ilo_lfs19

Number of strata =  26                        Number of obs    =      5,888
Number of PSUs   = 146                        Population size = 101,363.92

Subpop. no. PSUs = 146                        Subpop. no. obs =      4,627
                                              Subpop. size    = 79,968.717
                                              Design df       =        120

--------------------------------------------------------------------------------
                         |             Linearized            Logit
                         | Proportion   std. err.     [95% conf. interval]
-------------------------+------------------------------------------------------
               ilo_lfs19 |
           1 - Employed  |   .6556519   .0104248      .6347285    .6759887
         2 - Unemployed  |   .0224212      .0023      .0182919    .0274566
3 - Outside Labour Force |   .3219269   .0099233      .3026022    .3418807
--------------------------------------------------------------------------------
```

1. 4,627 working-age persons, standing for 79,969
2. 65.6% employed
3. 2.2% unemployed
4. 32.2% outside the labour force

# A proportion is a share between 0 and 1.

Multiply by 100 for the release: .656 is 65.6%.

**Keep the 0–1 scale in Stata; format at the end.**

# The unemployment rate:

```
svy: ratio unemp/lf
```



# The definition behind the ratio

Unemployed: without work, available for work, and seeking work.

Labour force: employed + unemployed.

ILO, Resolution concerning statistics of work, employment and labour underutilization, 19th ICLS (2013). `ilo_lfs19` already applies it.

YOUR TURN · TASK 17

# The national unemployment rate

**DO**

Type:

```
svy: ratio unemp/lf
```

**YOU SHOULD SEE**

One row `_ratio_1`: about .033, with SE and interval.

**IF NOT**

Red text? The slash has no spaces: `unemp/lf`.

**EXTRA** Express the rate and the interval in percent, one decimal.

# Read svy: ratio

```
. svy: ratio unemp/lf

Number of strata =  26            Number of obs    =      5,888
Number of PSUs   = 146            Population size = 101,363.92
                                  Design df       =        120

     _ratio_1: unemp/lf (1)

--------------------------------------------------------------
             |             Linearized
             |      Ratio   std. err.      [95% conf. interval]
-------------+------------------------------------------------
    _ratio_1 |    .033066   .0034775       .0261808    .0399513 (2)
--------------------------------------------------------------
```

1. Stata names the ratio `_ratio_1`
2. unemployment rate 3.3%; 95% CI 2.6% to 4.0%

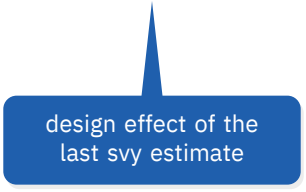
# A space works as well as the slash

```
. svy: ratio unemp lf
------------------------------------------------------------
             |             Linearized
             |      Ratio   std. err.     [95% conf. interval]
-------------+----------------------------------------------
    _ratio_1 |    .033066   .0034775      .0261808    .0399513  (1)
------------------------------------------------------------
```

(1) same ratio, same SE: Stata reads `unemp lf` as unemp over lf

# The price of the design:

```
estat effects
```



YOUR TURN · TASK 18

# Design effect of the rate

**DO**

Type, right after Task 16:

```
estat effects
```

**YOU SHOULD SEE**

The rate and SE again, with DEFF and DEFT.

**IF NOT**

`last estimates not found`? Run `svy: ratio unemp/lf` first.

# Read estat effects

```
------------------------------------------------------------
             |             Linearized
             |      Ratio   std. err.       DEFF       DEFT
-------------+----------------------------------------------
    _ratio_1 |    .033066   .0034775     1.19117   1.09141
------------------------------------------------------------
```

1

1 DEFF 1.19: the variance is 19% larger than in a simple random sample of the same size

# DEFF: how much the design inflates the variance.

1 = as good as simple random sampling. 2 = you need twice the interviews for the same precision.

**Clustering pushes it up; stratification pulls it down.**

# Each indicator has its own design effect

```
------------------------------------------------------------
             |             Linearized
             |      Mean   std. err.       DEFF       DEFT
-------------+----------------------------------------------
          lf |   .6780731   .0099233     2.09513   1.44746  (1)
------------------------------------------------------------
```

(1) participation rate: DEFF 2.1, the price of clustering is higher

# Hours cluster even more

```
. estat effects

------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.       DEFF       DEFT
-------------+----------------------------------------------
~1_how_usual |   39.62338   .4030652     3.56171   1.88725
------------------------------------------------------------
```

1

1. usual hours: DEFF 3.6. Neighbours work similar weeks

ON YOUR OWN · E

# The participation rate

**THE QUESTION**

Estimate the labour force participation rate: the share of the working-age population in the labour force.

**HINT**

A mean of the 0/1 flag `lf`, for the 15+ only: `subpop(wap)`.

ON YOUR OWN · E · ANSWER

```
svy, subpop(wap): mean lf
```

```
. svy, subpop(wap): mean lf

Number of strata =  26            Number of obs    =      5,888
Number of PSUs   = 146            Population size = 101,363.92
Subpop. no. PSUs = 146            Subpop. no. obs =      4,627
                                  Subpop. size    = 79,968.717
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
          lf |   .6780731   .0099233      .6584257    .6977206  (1)
--------------------------------------------------------------
```

(1) participation 67.8%; 95% CI 65.8% to 69.8%

# Two more headline indicators

```
. svy, subpop(wap): mean emp

Number of strata =  26            Number of obs   =      5,888
Number of PSUs   = 146            Population size = 101,363.92
Subpop. no. PSUs = 146            Subpop. no. obs =      4,627
                                  Subpop. size    = 79,968.717
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         emp |   .6556519   .0104248      .6350116    .6762922
--------------------------------------------------------------
```

1

1. employment-to-population ratio 65.6%

# Working-age men and women

```
. svy: total wap, over(ilo_sex)

Number of strata =  26             Number of obs    =      5,888
Number of PSUs   = 146             Population size = 101,363.92
                                   Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |      Total   std. err.     [95% conf. interval]
-------------+------------------------------------------------
c.wap@ilo_sex |
    1 - Male  |   37185.58   1151.712      34905.27    39465.89
  2 - Female  |   42783.13   1248.177      40311.83    45254.44
--------------------------------------------------------------
```

1. 37,186 men aged 15+
2. 42,783 women aged 15+

# Unemployed women and men

```
. svy: total unemp, over(ilo_sex)

Number of strata =  26               Number of obs   =       5,888
Number of PSUs   = 146               Population size = 101,363.92
                                     Design df       =         120

------------------------------------------------------------------
                |             Linearized
                |      Total   std. err.     [95% conf. interval]
----------------+-------------------------------------------------
c.unemp@ilo_sex |
      1 - Male  |   896.8056   156.7402      586.4709     1207.14  1
    2 - Female  |   896.1888   128.7942      641.1853    1151.192  2
------------------------------------------------------------------
```

1. 897 unemployed men
2. 896 unemployed women: the same number, intervals of about 600

# The working-age population

```
. svy: total wap

Number of strata =  26            Number of obs   =      5,888
Number of PSUs   = 146            Population size = 101,363.92
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |      Total   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         wap |   79968.72   2092.451      75825.81    84111.62 (1)
--------------------------------------------------------------

.
```

(1) 79,969 persons aged 15 and over

QUICK CHECK

## DEFF = 2.1 for the participation rate. What does it mean?

A. the rate is twice too high

B. the variance is about twice that of a simple random sample

C. two PSUs per stratum

ANSWER

## DEFF = 2.1 for the participation rate. What does it mean?

A. the rate is twice too high

**B. the variance is about twice that of a simple random sample**

C. two PSUs per stratum

The estimate is fine; its variance is 2.1 times what 4,627 independent interviews would give.

QUICK CHECK

## `svy: total emp` gives 52,432. What is that?

A. the number of employed interviews

B. the number of employed people in Seychelles

C. the share employed

ANSWER

## `svy: total emp` gives 52,432. What is that?

A. the number of employed interviews

**B. the number of employed people in Seychelles**

C. the share employed

A total of a 0/1 flag adds the weights of the rows with 1: people, not interviews (3,012 interviews).

# Part 4 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `generate f = (condition)` | a 0/1 flag, never missing |
| `svy: mean v` | design-based mean |
| `svy: total f` | number of people with f = 1 |
| `svy, subpop(wap): proportion v` | shares among the 15+ |
| `svy: ratio unemp/lf` | unemployment rate |
| `estat effects` | DEFF and DEFT of the last estimate |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 4**

| | | |
|---|---|---|
| CORE | P1 | Design-based mean age. Type it, two decimals. |
| CORE | P2 | Number of employed persons. Type the lower bound of the interval, rounded. |
| CORE | P3 | Number of unemployed persons, rounded. |
| EXTENSION | P4 | Employment-to-population ratio, in percent, one decimal. |
| CHALLENGE | P5 | Design effect of the participation rate, two decimals. |

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# You can produce NBS’s headline labour indicators with correct margins of error.

lost? open `catchup_D3_S4.do`



5

DAY 3 · PART 5

# Estimates for part of the population

THE DIRECTOR'S QUESTIONS, PART 2

# Regions, women, young people

**BY REGION**

Unemployment rate in each of the seven regions

**BY SEX**

Participation of women and men

**YOUTH**

Unemployment among 15–24 year olds

Each estimate is for part of the population. The design must stay whole.

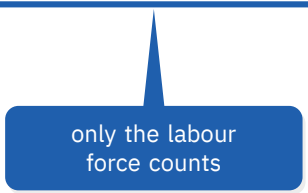
# Part 5: groups, the correct way

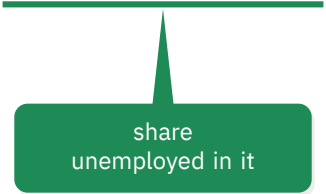
**AFTER THIS PART YOU CAN**

- estimate a rate for a group with `subpop()`
- explain why `if` can give a wrong standard error
- produce one estimate per region with `over()`
- read a small region's interval before its rate

# The unemployment rate, a second way:

```
svy, subpop(lf): mean unemp
```



YOUR TURN · TASK 19

# The rate with subpop()

**DO**

Type:

```
svy, subpop(lf): mean unemp
```

**YOU SHOULD SEE**

The same .033066 and the same SE as the ratio.

**IF NOT**

`variable lf not found`? Build the flags first (Task 12).

# Same rate, same SE

```
. svy, subpop(lf): mean unemp

Number of strata =  26            Number of obs    =      5,888
Number of PSUs   = 146            Population size = 101,363.92
Subpop. no. PSUs = 146            Subpop. no. obs =      3,126  (1)
                                  Subpop. size    = 54,224.639
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513  (2)
--------------------------------------------------------------
```

1. 3,126 in the labour force, 146 PSUs kept
2. .033066, SE .0034775: identical to the ratio

# subpop() after the comma works too

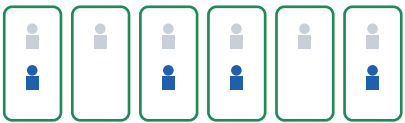
```
. svy: mean unemp, subpop(1f)
------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+----------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513
------------------------------------------------------------
```

1

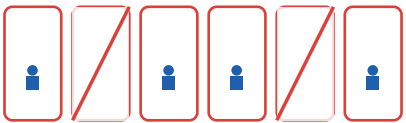
1. same rate, same SE as in the prefix

# What subpop() keeps, what if throws away

**`subpop()`: all PSUs stay**



**`if`: PSUs with no group member vanish**



Blue: in the group. Grey: outside. Stata's variance uses the PSUs; `if` removes those without a group member, so the variance is computed on a smaller design.

# Labour force: if and subpop() agree here

**if lf == 1**

```
. svy: mean unemp if lf == 1

------------------------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513   (1)
------------------------------------------------------------------------------
```

**subpop(lf)**

```
. svy, subpop(lf): mean unemp

------------------------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513   (2)
------------------------------------------------------------------------------
```

(1) (2) same SE: every PSU holds labour force members, so `if` drops none.

# Young people: if loses 21 PSUs

**if youth_lf == 1**

```
. svy: mean unemp if youth_lf == 1

Number of PSUs   = 125             Population size = 6,352.5252
                                   Design df       =         99
                 |             Linearized
                 |       Mean   std. err.     [95% conf. interval]
-----------------+------------------------------------------------
          unemp |   .1277627   .0212058      .0856858    .1698397
-----------------------------------------------------------------
```

1 2

**subpop(youth_lf)**

```
. svy, subpop(youth_lf): mean unemp

Number of PSUs   = 146             Population size = 101,363.92
                                   Design df       =         120
                 |             Linearized
                 |       Mean   std. err.     [95% conf. interval]
-----------------+------------------------------------------------
          unemp |   .1277627   .0212881      .0856137    .1699118
-----------------------------------------------------------------
```

3 4

1 125 PSUs, 99 df 2 SE .02121 3 146 PSUs, 120 df 4 SE .02129. Same rate; with `if` the variance is computed on a smaller design.

# Praslin and La Digue: identical, and why

**if lf == 1 & region == 6**

```
. svy: mean unemp if lf == 1 & region == 6

Number of strata =  3             Number of obs   =         280
                                  Design df       =          16

             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |   .0083458   .0042431     -.0006492    .0173408
-------------------------------------------------------------
```

1

**subpop(if lf == 1 & region == 6)**

```
. svy, subpop(if lf == 1 & region == 6): mean unemp
Number of strata =  3             Number of obs   =         514
                                  Design df       =          16

             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |   .0083458   .0042431     -.0006492    .0173408
-------------------------------------------------------------

Note: 23 strata omitted because they contain no subpopulation
      members.
```

2

1 3 strata, 16 df in both. 2 Strata without any group member add nothing to the variance, so Stata leaves them out either way. A region is a set of whole strata.

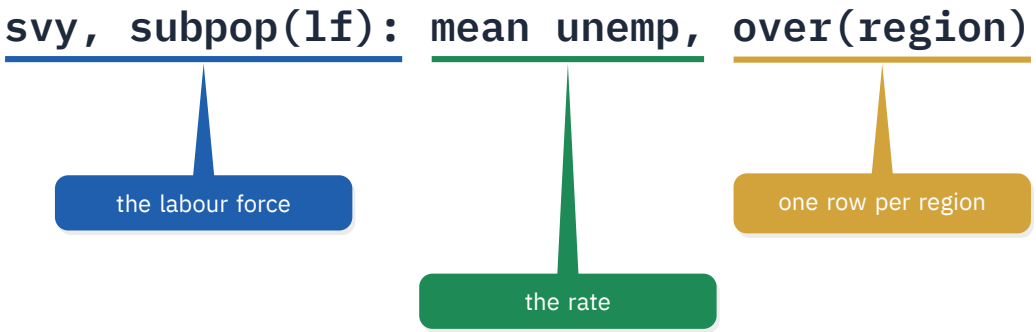
RULE

# Never `svy: mean ... if group`.

The rate is right, but the variance is computed on the PSUs that happen to hold group members: the design shrinks, and so do the degrees of freedom.

**Always `svy, subpop(group): ...`**

# One estimate per region:



YOUR TURN · TASK 20

# Unemployment rate by region

**DO**

Type:

```
svy, subpop(lf): mean unemp, over(region)
```

**YOU SHOULD SEE**

Seven rows `c.unemp@region`, each with SE and interval.

**IF NOT**

Only one row? Check the comma before `over()`.

**EXTRA** Which region has the widest interval?

# Read the regional rates

```
. svy, subpop(lf): mean unemp, over(region)

------------------------------------------------------------------------------
                             |             Linearized
                             |       Mean   std. err.     [95% conf. interval]
-----------------------------+------------------------------------------------
       c.unemp@region        |
            1- North         |   .0320638   .0085409      .0151535    .0489741
          2- Central 1       |   .0453336   .0069975       .031479    .0591883
          3- Central 2       |   .0361783   .0110366      .0143267       .05803
          4- EastSouth       |   .0267923   .0061746      .0145671    .0390175
               5- West       |   .0531623   .0143954      .0246605    .0816642
6- Praslin & LaDigue         |   .0083458   .0042431     -.0000553    .0167469
     7- Perseverance         |   .0162846   .0077196      .0010003    .0315688
------------------------------------------------------------------------------
```

1. Central 1: 4.5% (3.1 to 5.9)
2. West: 5.3%, the widest interval (2.5 to 8.2)
3. Praslin & La Digue: 0.8%, and the lower bound is below zero

# A small rate needs a different interval

**mean: symmetric interval**

```
                        |            Linearized
                        |      Mean   std. err.     [95% conf. interval]
------------------------+------------------------------------------------
6- Praslin & LaDigue    |   .0083458   .0042431     -.0000553    .0167469
```

1

**proportion: logit interval**

```
. svy, subpop(if lf == 1 & region == 6): proportion unemp

------------------------------------------------------------------------------
             |             Linearized            Logit
             | Proportion   std. err.     [95% conf. interval]
-------------+----------------------------------------------------------------
       unemp |
          0  |   .9916542   .0042431      .9756543    .9971695
          1  |   .0083458   .0042431      .0028305    .0243457
------------------------------------------------------------------------------
Note: 23 strata omitted because they contain no subpopulation
      members.
```

2

1 0.8%, from below zero to 1.7%. 2 0.8%, from 0.28% to 2.43%: a logit interval cannot go below zero.

YOUR TURN · TASK 21

# Participation by region and by sex

**DO**

Type:

```
svy, subpop(wap): mean lf, over(region)
svy, subpop(wap): mean lf, over(ilo_sex)
```

**YOU SHOULD SEE**

Seven regional rates, then two: men and women.

**IF NOT**

`variable lf not found`? Build the flags (Task 12).

**EXTRA** Which region has the lowest participation?

# Participation by region

```
. svy, subpop(wap): mean lf, over(region)

------------------------------------------------------------------------------
                     |             Linearized
                     |       Mean   std. err.     [95% conf. interval]
---------------------+--------------------------------------------------------
          c.lf@region |
            1- North |   .7126907   .0242906      .6645971    .7607843
       2- Central 1  |    .686534   .0213907      .6441819    .7288861
       3- Central 2  |   .5863703   .0170308      .5526506    .6200901
       4- EastSouth  |   .6534159   .0141036      .6254918      .68134
             5- West |   .7400635   .0393773      .6620992    .8180278
6- Praslin & LaDigue |   .6887712   .0247656      .6397371    .7378054
     7- Perseverance |   .6695755   .0224952      .6250366    .7141143
------------------------------------------------------------------------------
```

1. Central 2: 58.6%, the lowest
2. West: 74.0%, the highest, with the widest interval

# Participation by sex

```
. svy, subpop(wap): mean lf, over(ilo_sex)

------------------------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+----------------------------------------------------------------
c.lf@ilo_sex |
   1 - Male  |   .6980329   .0117153      .6748375    .7212283   1
 2 - Female  |   .6607248    .012067      .6368331    .6846166   2
------------------------------------------------------------------------------
```

1. men 69.8% (67.5 to 72.1)
2. women 66.1% (63.7 to 68.5)

ON YOUR OWN · F

# Unemployment of women and men

**THE QUESTION**

Estimate the unemployment rate separately for women and men. Is the difference clear?

**HINT**

Same command as the regions, with `over(ilo_sex)`.

**ON YOUR OWN · F · ANSWER**

```
svy, subpop(lf): mean unemp, over(ilo_sex)
```

```
. svy, subpop(lf): mean unemp, over(ilo_sex)

------------------------------------------------------------------------------
                |             Linearized
                |       Mean   std. err.      [95% conf. interval]
----------------+-------------------------------------------------------------
c.unemp@ilo_sex |
      1 - Male  |     .03455   .0061649      .0223439    .0467561   (1)
    2 - Female  |   .0317034   .0045353      .0227238     .040683   (2)
------------------------------------------------------------------------------
```

1. men 3.5%
2. women 3.2%: the intervals overlap almost entirely: no clear difference

# Youth unemployment

```
. svy, subpop(if lf == 1 & agegrp == 1): mean unemp

Number of strata =  26            Number of obs    =      5,888
Number of PSUs   = 146            Population size = 101,363.92
Subpop. no. PSUs = 125            Subpop. no. obs =        361   1
                                  Subpop. size    = 6,352.5252
                                  Design df       =        120

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
       unemp |   .1277627   .0212881      .0856137    .1699118   2
--------------------------------------------------------------
```

1. 361 young people in 125 of the 146 PSUs
2. 12.8%, four times the national rate; CI 8.6% to 17.0%

# Two groups at once

```
. svy, subpop(lf): mean unemp, over(ilo_sex agegrp)

------------------------------------------------------------------------------
                     |             Linearized
                     |       Mean   std. err.     [95% conf. interval]
---------------------+--------------------------------------------------------
c.unemp@ilo_sex#agegrp |
      1 - Male#15-24 |   .1412905   .0302204      .0814563    .2011247
      1 - Male#25-34 |   .0288019   .0111334      .0067586    .0508453
      1 - Male#35-54 |   .0220637   .0070503      .0081047    .0360227
        1 - Male#55+ |   .0121354   .0065619     -.0008567    .0251275
    2 - Female#15-24 |    .116769   .0292666      .0588231    .1747149
    2 - Female#25-34 |   .0313583   .0093892      .0127683    .0499483
    2 - Female#35-54 |   .0171527   .0047713      .0077058    .0265996
      2 - Female#55+ |    .013974   .0075769     -.0010277    .0289757
------------------------------------------------------------------------------

.
```

1. young men 14.1%
2. young women 11.7%
3. men 55+: 1.2%, interval below zero

COMMON ERROR

# over() with an identifier

```
. svy, subpop(lf): mean unemp, over(unique_ind)
(running mean on estimation sample)
string variables not allowed in varlist;
unique_ind is a string variable
invalid over() option
```

**WHY**

`over()` needs a numeric group variable. `unique_ind` is text, and one group per person would make no sense.

**FIX**

Use a categorical numeric variable: `region`, `ilo_sex`, `agegrp`.

# subpop() with a variable that is not 0/1

```
. svy, subpop(region): mean unemp

Note: subpop() takes on values other than 0 and 1 (1)
-------------+------------------------------------------------
       unemp |   .0176887   .0018004      .0141241    .0212533 (2)
-------------------------------------------------------------
```

1. only a note: any non-zero value counts as in the group
2. every region is non-zero: the whole population

# Employment-to-population ratio by region

```
. svy, subpop(wap): mean emp, over(region)

------------------------------------------------------------------------------
                     |             Linearized
                     |       Mean   std. err.      [95% conf. interval]
---------------------+--------------------------------------------------------
         c.emp@region |
            1- North  |   .6898391   .0259848       .638391    .7412872
        2- Central 1  |   .6554109   .0215099      .6128228    .6979991
        3- Central 2  |   .5651564   .0160602      .5333584    .5969545
        4- EastSouth  |   .6359094   .0135007       .609179    .6626398
             5- West  |    .70072    .0447723      .6120739    .7893661
6- Praslin & LaDigue  |   .6830229   .0246058      .6343051    .7317407
     7- Perseverance  |   .6586717   .0231664      .6128039    .7045395
------------------------------------------------------------------------------
```

1. Central 2: 56.5%, the lowest
2. West: 70.1%

# A group flag with missing values

```
. svy, subpop(ilo_joball_tru): mean emp

Number of strata = 19              Number of obs    =        57
Number of PSUs   = 35              Population size = 834.93344
Subpop. no. PSUs = 35              Subpop. no. obs =        57
                                   Subpop. size    = 834.93344
                                   Design df       =        16

--------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.     [95% conf. interval]
-------------+------------------------------------------------
         emp |          1          .              .           .
--------------------------------------------------------------
Note: Missing standard error because of stratum with single
      sampling unit.
```

1. only the 57 underemployed rows are left: 19 strata, 35 PSUs
2. no standard error at all
3. some stratum has a single PSU left

Praslin & La Digue has few interviews: read its interval before its rate.

QUICK CHECK

## Why `subpop()` and not `if`?

A. subpop is faster

B. subpop keeps all PSUs in the variance calculation

C. if gives a different rate

ANSWER

## Why `subpop()` and not `if`?

A subpop is faster

**B** subpop keeps all PSUs in the variance calculation

C if gives a different rate

The rate is the same either way; the standard error is only right when the whole design is kept.

QUICK CHECK

## What does `svy, subpop(region): mean unemp` estimate?

A. nothing: it stops with an error

B. the rate for the whole population: every region code is non-zero

C. one rate per region

ANSWER

## What does `svy, subpop(region): mean unemp` estimate?

A nothing: it stops with an error

**B** the rate for the whole population: every region code is non-zero

C one rate per region

Log P09: a note, then the whole population. Use a 0/1 flag, or `over(region)` for one rate per region.

# Part 5 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `svy, subpop(f): mean y` | estimate for the group f = 1 |
| `svy, subpop(if a == 1 & b == 2): ...` | group built inline |
| `..., over(region)` | one estimate per category |
| `..., over(sex agegrp)` | every combination |
| `svy: proportion f (subpop)` | interval that stays in 0-1 |
| `never: svy: mean y if group` | wrong SE for small groups |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 5**

CORE P1 Participation rate of women, percent, one decimal.

CORE P2 Region with the highest unemployment rate: type its rate in percent.

CORE P3 Youth (15–24) unemployment rate, percent, one decimal.

EXTENSION P4 Unemployment rate of young women, percent, one decimal.

CHALLENGE P5 Employment-to-population ratio by region: type the lowest, percent.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# You can produce regional labour indicators that respect the design.

lost? open `catchup_D3_S5.do`



6

DAY 3 · PART 6

# Weighted tables

# Part 6: survey tables

**AFTER THIS PART YOU CAN**

- produce a one-way weighted table with `svy: tabulate`
- show counts, percentages, SEs and intervals
- produce a two-way table with row or column percentages
- read the design-based test under the table

# Yesterday's table, for Seychelles:



YOUR TURN · TASK 22

# One-way survey table

**DO**

Type:

```
svy: tabulate region
```

**YOU SHOULD SEE**

Seven shares that add up to 1.

**IF NOT**

`data not set up for svy`? Run the svyset line or open `lfs_svy.dta`.

# Read svy: tabulate

```
. svy: tabulate region

----------------------
   Region | proportion
----------+-----------
 1- North |       .179
 2- Centr |       .127
 3- Centr |      .1206
 4- EastS |      .2547
  5- West |      .1524
 6- Prasl |      .1099
 7- Perse |      .0563
          |
    Total |          1
----------------------
Key: proportion = Cell proportion
```

1. EastSouth holds 25.5% of the population
2. by default: cell proportions; labels cut to 8 characters

# Options for the release

```
svy: tabulate region, count se ci format(%9.0fc)
svy: tabulate region, percent ci format(%5.1f)
```

- `count`: weighted numbers of people
- `percent`: percentages instead of proportions
- `se`, `ci`: standard error, interval
- `format()`: how the numbers are shown

YOUR TURN · TASK 23

# Counts and percentages with intervals

**DO**

Type both lines:

```
svy: tabulate region, count se ci format(%9.0fc)
svy: tabulate region, percent ci format(%5.1f)
```

**YOU SHOULD SEE**

Two tables: people with SE and bounds, then percentages with bounds.

**IF NOT**

`invalid format`? Copy the format exactly: `%9.0fc`, `%5.1f`.

# Read the counts

```
. svy: tabulate region, count se ci format(%9.0fc)

------------------------------------------------------------
   Region |      count          se          lb          ub
----------+-------------------------------------------------
 1- North |     18,147         864      16,437      19,857
 2- Centr |     12,876         783      11,325      14,427
 3- Centr |     12,229         898      10,451      14,006
 4- EastS |     25,821       1,573      22,707      28,934
  5- West |     15,444       1,021      13,423      17,464
 6- Prasl |     11,139         843       9,470      12,808
 7- Perse |      5,709         742       4,240       7,179
          |
    Total |    101,364
------------------------------------------------------------
Key:    count = Weighted count
          se = Linearized standard error of weighted count
          lb = Lower 95% confidence bound for weighted count
          ub = Upper 95% confidence bound for weighted count
```

1. North: 18,147 people (16,437 to 19,857)
2. total 101,364: no interval, it is fixed by the weights

# Read the percentages

```
. svy: tabulate region, percent ci format(%5.1f)

-----------------------------------------------
   Region | percentage          lb          ub
----------+------------------------------------
 1- North |       17.9        16.3        19.6
 2- Centr |       12.7        11.3        14.3
 3- Centr |       12.1        10.5        13.8
 4- EastS |       25.5        23.0        28.1
  5- West |       15.2        13.5        17.2
 6- Prasl |       11.0         9.5        12.6
 7- Perse |        5.6         4.4         7.2
          |
    Total |      100.0
-----------------------------------------------
Key: percentage = Cell percentage
             lb = Lower 95% confidence bound for cell percentage
             ub = Upper 95% confidence bound for cell percentage
```

1

1 Perseverance: 5.6% of the population (4.4 to 7.2)

YOUR TURN · TASK 24

# Two-way table, row percentages

**DO**

Type:

```
svy: tabulate region ilo_lfs19, row percent
format(%5.1f)
```

**YOU SHOULD SEE**

Status within each region, each row adding to 100, and a Pearson test below.

**IF NOT**

`only one of options ...`? Remove `se` or `count` (see the error slide).

# Read the two-way table

```
. svy: tabulate region ilo_lfs19, row percent format(%5.1f)

-------------------------------------------------------------
          |   Labour Force Status - 19th ICLS
   Region | 1 - Empl  2 - Unem  3 - Outs     Total
----------+--------------------------------------------------
 1- North |     69.0       2.3      28.7     100.0
 2- Centr |     65.5       3.1      31.3     100.0

 3- Centr |     56.5       2.1      41.4     100.0

 4- EastS |     63.6       1.8      34.7     100.0
  5- West |     70.1       3.9      26.0     100.0
 6- Praml |     68.3       0.6      31.1     100.0
 7- Perse |     65.9       1.1      33.0     100.0
          |

    Total |     65.6       2.2      32.2     100.0
-------------------------------------------------------------
Key: Row percentage

  Pearson:
    Uncorrected   chi2(12)        =   61.2046

    Design-based  F(6.06, 727.14) =    3.4750     P = 0.0021
```

1. Central 2: 41.4% outside the labour force
2. Seychelles: 65.6 / 2.2 / 32.2
3. design-based test: status differs between regions (P = 0.002)

# Sample or Seychelles?

**YESTERDAY: tabulate, row**

| Region | 1 - Emplo | 2 - Unemp | 3 - Outsi | Total |
|---|---|---|---|---|
| 1- North | 69.78 | 2.34 | 27.88 | 100.00 |
| 2- Central 1 | 64.71 | 3.50 | 31.79 | 100.00 |

Labour Force Status - 19th ICLS

**TODAY: svy: tabulate, row**

| Region | 1 - Empl | 2 - Unem | 3 - Outs | Total |
|---|---|---|---|---|
| 1- North | 69.0 | 2.3 | 28.7 | 100.0 |
| 2- Centr | 65.5 | 3.1 | 31.3 | 100.0 |

Labour Force Status - 19th ICLS

1. 3.50% of Central 1's interviews are unemployed.
2. 3.1% of its population is.

ON YOUR OWN · G

# Status by sex, for Seychelles

**THE QUESTION**

Produce the weighted row percentages of labour force status for women and men. Is the difference significant?

**HINT**

Two variables in `svy: tabulate`, option `row`.

**ON YOUR OWN · G · ANSWER**

```
svy: tabulate ilo_sex ilo_lfs19, row percent
format(%5.1f)
```

```
. svy: tabulate ilo_sex ilo_lfs19, row percent format(%5.1f)

-------------------------------------------------
          |    Labour Force Status - 19th ICLS
      Sex | 1 - Empl  2 - Unem  3 - Outs     Total
----------+--------------------------------------
 1 - Male |     67.4       2.4      30.2     100.0
 2 - Fema |     64.0       2.1      33.9     100.0
          |
    Total |     65.6       2.2      32.2     100.0
-------------------------------------------------
Key: Row percentage

  Pearson:
    Uncorrected   chi2(2)         =    7.5387
    Design-based  F(1.85, 221.67) =    2.9922     P = 0.0564
```

1. women: 64.0% employed, 33.9% outside
2. P = 0.056: not significant at 5%

COMMON ERROR

# Too many things at once

```
. svy: tabulate region ilo_lfs19, row count se ci percent
only one of options cell, count, row, or column can be specified when option se is specified
```

**WHY**

With `se`, the table can show only one kind of cell: row, column, cell or count.

**FIX**

Ask for one: `svy: tabulate region ilo_lfs19, row se`.

COMMON ERROR

# estat effects after a survey table

```
. estat effects
estat effects is not allowed after svy: tabulate
```

**WHY**

Design effects are available after svy estimation commands such as mean, total, proportion, ratio, not after svy: tabulate.

**FIX**

For a DEFF, estimate the same share with `svy: proportion` and run `estat effects`.

# Column percentages

```
. svy: tabulate region ilo_lfs19, column percent format(%5.1f)

-----------------------------------------------------------
                |    Labour Force Status - 19th ICLS
         Region | 1 - Empl  2 - Unem  3 - Outs      Total
----------------+------------------------------------------
 1- North       |     18.3      17.7      15.5       17.4
 2- Centr       |     13.0      18.0      12.6       13.0
 3- Centr       |     10.5      11.5      15.6       12.2
 4- EastS       |     25.0      20.1      27.7       25.7
                |
 5- West        |     16.7      27.3      12.6       15.6
 6- Prasl       |     11.4       2.8      10.6       11.0
 7- Perse       |      5.2       2.5       5.3        5.2
                |
          Total |    100.0     100.0     100.0      100.0
-----------------------------------------------------------
Key: Column percentage

  Pearson:
    Uncorrected   chi2(12)        =   61.2046
    Design-based  F(6.06, 727.14) =    3.4750     P = 0.0021

.
```

1

1 West: 27.3% of all the unemployed live there, 15.6% of the 15+

QUICK CHECK

## You want the share of the unemployed who live in each region. Which option?

A row

B column

C count

ANSWER

## You want the share of the unemployed who live in each region. Which option?

A row

**B** column

C count

Columns are the statuses: `column` makes each status add up to 100 across the regions.

QUICK CHECK

## Under `svy: tabulate a b`, which test do you report?

A. Uncorrected chi2

B. Design-based F, with its P

C. none: weighted tables have no test

ANSWER

## Under `svy: tabulate a b`, which test do you report?

A. Uncorrected chi2

**B. Design-based F, with its P**

C. none: weighted tables have no test

The uncorrected chi2 treats the interviews as a simple random sample.

# Part 6 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `svy: tabulate v` | weighted proportions |
| `..., count se ci format(%9.0fc)` | people, SE, interval |
| `..., percent ci format(%5.1f)` | percentages with interval |
| `svy: tabulate a b, row` | b within each a |
| `svy: tabulate a b, column` | a within each b |
| `Design-based F, P` | the test to read |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 6**

CORE P1 Weighted percentage of women among the 15+ (`svy, subpop(wap): proportion`).

CORE P2 Percentage of the population in Perseverance, with its interval.

CORE P3 Weighted row percentages of status by region: type the unemployed % in West.

EXTENSION P4 Column percentages: share of the unemployed living in North.

CHALLENGE P5 Weighted count of people in Central 2, with SE.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# Yesterday’s sample tables are now population tables.

lost? open **catchup_D3_S6.do**



7

DAY 3 · PART 7

# Good enough to publish?

# Part 7: precision and the five errors

**AFTER THIS PART YOU CAN**

- compute the coefficient of variation with `estat cv`
- see how many interviews stand behind each estimate
- recognise the five survey errors that reach publications

# The coefficient of variation:

```
estat cv
```

SE / estimate,
in percent, for
each estimate

YOUR TURN · TASK 25

# CV of the regional rates

**DO**

Type:

```
svy, subpop(lf): mean unemp, over(region)
estat cv
```

**YOU SHOULD SEE**

The seven rates again, with a CV column in percent.

**IF NOT**

`invalid subcommand cv`? The last estimate was not `svy:` (next slides).

# Read estat cv

```
. estat cv

-----------------------------------------------
             |             Linearized
        Over |       Mean   std. err.      CV (%)
-------------+---------------------------------
     c.unemp@|
      region |
   1- North   |   .0320638   .0085409     26.6371
2- Centra..   |   .0453336   .0069975     15.4357
3- Centra..   |   .0361783   .0110366      30.506
4- EastSo..   |   .0267923   .0061746     23.0461
    5- West   |   .0531623   .0143954     27.0782
6- Prasli..   |   .0083458   .0042431     50.8416
7- Persev..   |   .0162846   .0077196     47.4042
-----------------------------------------------
```

1. Central 1: CV 15%
2. Praslin & La Digue: 51%
3. Perseverance: 47%

# Flag or suppress estimates whose CV is above NBS's threshold.

Many offices flag above 15–20% and suppress above 30–35%. NBS sets its own rule.

**Apply NBS's threshold (open item 10).**

# How many interviews behind each estimate

```
. estat size

------------------------------------------------------------------------------
             |             Linearized
        Over |       Mean   std. err.              Obs             Size
-------------+----------------------------------------------------------------
     c.unemp@|
      region |
   1- North   |   .0320638   .0085409              525   9,907.7850094
2- Centra..   |   .0453336   .0069975              487   7,113.9764128
3- Centra..   |   .0361783   .0110366              399   5,707.9342985
4- EastSo..   |   .0267923   .0061746              765   13,450.365383
    5- West   |   .0531623   .0143954              492   9,222.8042846
6- Prasli..   |   .0083458   .0042431              280   6,049.1918354
7- Persev..   |   .0162846   .0077196              178   2,772.5813437 (1)
------------------------------------------------------------------------------
```

(1) Perseverance: 178 interviews in the labour force, standing for 2,773 people

COMMON ERROR

# estat cv after a plain command

```
. estat cv
invalid subcommand cv
```

**WHY**

`estat cv` works only after `svy:`. The last estimate was a plain `mean`.

**FIX**

Run the `svy:` estimate again, then `estat cv`.

# Error 1: aweight instead of svy:

**mean ... [aweight] if lf == 1**

```
. mean unemp [aweight = wgt_final] if lf == 1

------------------------------------------------------------------------------
             |       Mean   Std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
       unemp |    .033066   .0031986       .0267944    .0393377
------------------------------------------------------------------------------
```

1

**svy, subpop(lf): mean**

```
. svy, subpop(lf): mean unemp

------------------------------------------------------------------------------
             |             Linearized
             |       Mean   std. err.      [95% conf. interval]
-------------+----------------------------------------------------------------
       unemp |    .033066   .0034775       .0261808    .0399513
------------------------------------------------------------------------------
```

2

1 2 same rate. SE .0032 against .0035: the weighted plain mean is 8% too optimistic.

# Error 3: forgetting the strata

**svyset psu [pw = wgt_final]**

```
. svy, subpop(lf): mean unemp
Number of strata =   1            Number of obs   =      5,888
                   |             Linearized
                   |       Mean   std. err.     [95% conf. interval]
-------------------+------------------------------------------------
       unemp |    .033066   .0037572      .0256402    .0404919
-------------------------------------------------------------------
```

1

**with strata(district)**

```
. svy, subpop(lf): mean unemp
Number of strata =  26            Number of obs   =      5,888
                   |             Linearized
                   |       Mean   std. err.     [95% conf. interval]
-------------------+------------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513
-------------------------------------------------------------------
```

2

1 1 stratum, SE .0038. 2 26 strata, SE .0035. Forgetting the strata overstates the error here.

# Error 4: the wrong weight

**pweight = wgt_base**

```
. svy, subpop(lf): mean unemp
Number of PSUs   = 146             Population size = 35,311.594
                 |             Linearized
                 |       Mean   std. err.     [95% conf. interval]
-----------------+------------------------------------------------
       unemp |   .0340644   .0035649      .0270062    .0411227
-------------------------------------------------------------------
```

1

**pweight = wgt_final**

```
. svy, subpop(lf): mean unemp
Number of PSUs   = 146             Population size = 101,363.92
                 |             Linearized
                 |       Mean   std. err.     [95% conf. interval]
-----------------+------------------------------------------------
       unemp |    .033066   .0034775      .0261808    .0399513
-------------------------------------------------------------------
```

3

1 the base weight stands for 35,312 people, not 101,364. 2 3.41% instead of 3 3.31%.

# Ignoring the clusters

```
. svy, subpop(lf): mean unemp
Number of PSUs   = 5,888          Population size = 101,363.92
              |             Linearized
              |       Mean   std. err.     [95% conf. interval]
--------------+-------------------------------------------------
       unemp |    .033066   .0034978      .0262091     .039923
-------------------------------------------------------------
```

1. each person treated as a PSU: 5,888 PSUs
2. SE .0035, close to the correct one for this rate

# Design effects differ by region

```
. estat effects

------------------------------------------------------------
             |             Linearized
      Over   |       Mean   std. err.       DEFF       DEFT
-------------+----------------------------------------------
     c.unemp@|
      region |
   1- North  |   .0320638   .0085409     1.35247   1.16296
2- Centra..  |   .0453336   .0069975     .467458   .683709
3- Centra..  |   .0361783   .0110366     1.15801   1.07611
4- EastSo..  |   .0267923   .0061746      1.1422   1.06874
    5- West  |   .0531623   .0143954     2.20516   1.48498
6- Prasli..  |   .0083458   .0042431     .764281   .874232
7- Persev..  |   .0162846   .0077196     .599012   .773959
------------------------------------------------------------

.
```

1 Central 1: DEFF 0.47: unemployment is spread evenly across its PSUs

2 West: DEFF 2.2: concentrated in some PSUs

# Design effects by age group

```
. estat effects

------------------------------------------------------------
             |             Linearized
        Over |       Mean   std. err.       DEFF       DEFT
-------------+----------------------------------------------
     c.unemp@|
      agegrp |
       15-24 |   .1277627   .0212881     1.50035    1.22489
       25-34 |   .0300666   .0068401     1.08697    1.04258
       35-54 |   .0194959   .0040849     1.23851    1.11289
         55+ |   .0131063   .0048821      1.2603    1.12263
------------------------------------------------------------
```

1

1 youth: DEFF 1.5. The other groups 1.1 to 1.3

# Time-related underemployment

```
. svy, subpop(emp): mean tru
        tru |   .0159242   .0030132      .0099583    .0218901
------------------------------------------------------------------------------
------------+-----------------------------------
        tru |   .0159242   .0030132      18.922
-----------------------------------------------
```

1

1 1.6% of the employed (1.0 to 2.2); CV 18.9%

# The five errors that reach publications

1. `[aweight]` or `[fweight]` instead of `svy:` right estimate, wrong SE
2. `if` instead of `subpop()`: wrong SE for small groups
3. forgetting `strata()`: SE changes
4. `wgt_base` or no weight: wrong estimate, wrong population
5. a regional rate published without its interval and CV

ON YOUR OWN · H

# Are the participation rates precise?

**THE QUESTION**

Compute the CV of the participation rate in each region. Is any of them above 10%?

**HINT**

Participation by region, then `estat cv`.

**ON YOUR OWN · H · ANSWER**

```
svy, subpop(wap): mean lf, over(region)
estat cv
```

```
. estat cv

-------------------------------------------------
             |             Linearized
      Over   |       Mean   std. err.      CV (%)
-------------+-----------------------------------
 c.lf@region |
   1- North  |   .7126907   .0242906      3.40829
2- Centra..  |    .686534   .0213907      3.11575
3- Centra..  |   .5863703   .0170308      2.90444
4- EastSo..  |   .6534159   .0141036      2.15844  (2)
    5- West  |   .7400635   .0393773       5.3208  (1)
6- Prasli..  |   .6887712   .0247656      3.59562
7- Persev..  |   .6695755   .0224952      3.35962
-------------------------------------------------
```

1. highest CV: West, 5.3%
2. lowest: EastSouth, 2.2%. All well below 10%

QUICK CHECK

## Praslin & La Digue: unemployment 0.8%, CV 51%. What do you publish?

A. the rate, one decimal

B. the rate flagged or suppressed under NBS's CV rule

C. the rate rounded to zero

ANSWER

## Praslin & La Digue: unemployment 0.8%, CV 51%. What do you publish?

A the rate, one decimal

**B** the rate flagged or suppressed under NBS's CV rule

C the rate rounded to zero

The number is too imprecise to print without a warning. NBS's threshold decides flag or suppress.

QUICK CHECK

## Without strata the SE is .0038; with strata .0035. Which do you publish?

A .0038, to be safe

B .0035: it comes from the design that was used

C the average of the two

ANSWER

## Without strata the SE is .0038; with strata .0035. Which do you publish?

A .0038, to be safe

**B .0035: it comes from the design that was used**

C the average of the two

The design decides the SE, not caution. Log C05.

# Part 7 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `estat cv` | CV in percent, after svy: |
| `estat size` | interviews and population per estimate |
| `estat effects` | DEFF and DEFT |
| `never [aweight] for a release` | wrong SE |
| `svyset with strata()` | or the SE changes |
| `pweight = wgt_final` | never wgt_base |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 7**

- CORE — P1 CV of the unemployment rate in Central 2.
- CORE — P2 CV of the participation rate in West.
- CORE — P3 Interviews behind the Perseverance unemployment rate (`estat size`).
- EXTENSION — P4 SE of the national rate without strata, four decimals.
- CHALLENGE — P5 Time-related underemployment among the employed: rate and CV.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# You know the five survey errors that most often reach a publication, and how to avoid each.

lost? open `catchup_D3_S7.do`

8

DAY 3 · PART 8

# Today’s product

# Part 8: the estimates table

**AFTER THIS PART YOU CAN**

- find every estimate, SE and bound in `r(table)`
- copy and reshape it with `matrix`
- write it to Excel with `putexcel ... matrix()`
- add a notes sheet with the design and the source

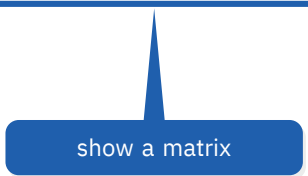
# After an estimate, the results live in `r(table)`.

One matrix: estimates, SEs, bounds, for every row of the output.

**Copy it before the next command replaces it.**

# See the result matrix:

```
matrix list r(table)
```






YOUR TURN · TASK 26

# List the result matrix

**DO**

Type:

```
svy, subpop(lf): mean unemp, over(region)
matrix list r(table)
```

**YOU SHOULD SEE**

A 9 x 7 matrix: rows b, se, t, pvalue, ll, ul, df, crit, eform; one column per region.

**IF NOT**

`matrix r(table) not found`? Run the two lines together, nothing in between.

# Read r(table)

```
. matrix list r(table)

r(table)[9,7]
          c.unemp@    c.unemp@    c.unemp@    c.unemp@    c.unemp@    c.unemp@    c.unemp@
          1.region    2.region    3.region    4.region    5.region    6.region    7.region
     b   .03206382   .04533362   .03617833   .02679227   .05316234    .0083458   .01628456
    se   .00854086   .00699755   .01103657   .00617457   .01439538   .00424313   .00771957
     t   3.7541685   6.4785011    3.27804   4.3391327   3.6930139   1.9668948   2.1095163
pvalue   .00026957   2.136e-09   .00136768       .00003   .00033492   .05150323   .03697815
    ll   .01515351   .03147896   .01432668   .01456705   .02466049  -.00005531   .00100035
    ul   .04897412   .05918827   .05802997   .03901748   .08166418    .0167469   .03156878
    df         120         120         120         120         120         120         120
  crit   1.9799304   1.9799304   1.9799304   1.9799304   1.9799304   1.9799304   1.9799304
 eform           0           0           0           0           0           0           0
```

1. b: the seven rates
2. se: their standard errors
3. ll, ul: the interval

# Copy it, turn it

```
matrix T = r(table)'
matrix list T
```

- `matrix T =`: a copy that survives the next command
- the apostrophe transposes: regions become rows, like the release

# Regions as rows

```
T[7,9]
                    b          se           t      pvalue          ll          ul          df        crit       eform
c.unemp@
1.region    .03206382   .00854086   3.7541685   .00026957   .01515351   .04897412         120   1.9799304           0  (1)
c.unemp@
2.region    .04533362   .00699755   6.4785011   2.136e-09   .03147896   .05918827         120   1.9799304           0
c.unemp@
3.region    .03617833   .01103657     3.27804   .00136768   .01432668   .05802997         120   1.9799304           0
c.unemp@
4.region    .02679227   .00617457   4.3391327       .00003   .01456705   .03901748         120   1.9799304           0
c.unemp@
5.region    .05316234   .01439538   3.6930139   .00033492   .02466049   .08166418         120   1.9799304           0
c.unemp@
6.region     .0083458   .00424313   1.9668948   .05150323   -.00005531   .0167469         120   1.9799304           0
c.unemp@
7.region    .01628456   .00771957   2.1095163   .03697815   .00100035   .03156878         120   1.9799304           0
```

(1) one row per region; columns b, se, t, pvalue, ll, ul ...

COMMON ERROR

# r(table) is gone

```
. matrix list r(table)
matrix r(table) not found
```

**WHY**

`count` is a new command: it replaced the stored results.

**FIX**

Copy with `matrix T = r(table)'` directly after the estimate.

COMMON ERROR

# estat cv replaces r(table) too

```
. matrix list r(table)
matrix r(table) not found
```

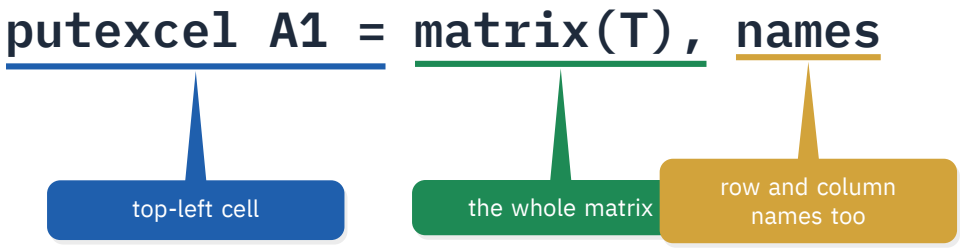
**WHY**

`estat cv` stores its own results.

**FIX**

Copy first, then `estat cv`.

# The matrix into Excel:



YOUR TURN · TASK 27

# The estimates sheet

**DO**

Run as one block:

```
svy, subpop(lf): mean unemp, over(region)
matrix T = r(table)'
putexcel set "output/estimates_q1.xlsx", ///
    sheet("Unemployment") replace
putexcel A1 = matrix(T), names
```

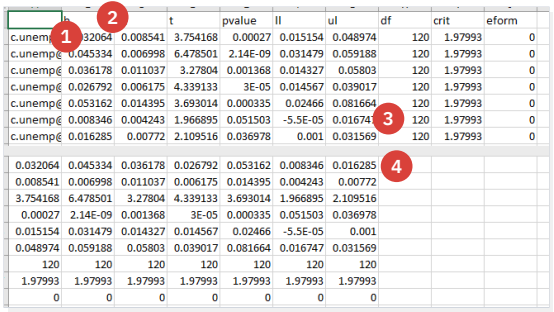
**YOU SHOULD SEE**

`file output/estimates_q1.xlsx saved`. Open it: seven rows, nine columns.

**IF NOT**

`file could not be saved`? Close Excel.

# The sheet in Excel



| | b | se | t | pvalue | ll | ul | df | crit | eform |
|---|---|---|---|---|---|---|---|---|---|
| c.unemp | 32064 | 0.008541 | 3.754168 | 0.00027 | 0.015154 | 0.048974 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.045334 | 0.006998 | 6.478501 | 2.14E-09 | 0.031479 | 0.059188 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.036178 | 0.011037 | 3.27804 | 0.001368 | 0.014327 | 0.05803 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.026792 | 0.006175 | 4.339133 | 3E-05 | 0.014567 | 0.039017 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.053162 | 0.014395 | 3.693014 | 0.000335 | 0.02466 | 0.081664 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.008346 | 0.004243 | 1.966895 | 0.051503 | -5.5E-05 | 0.016747 | 120 | 1.97993 | 0 |
| c.unemp@ | 0.016285 | 0.00772 | 2.109516 | 0.036978 | 0.001 | 0.031569 | 120 | 1.97993 | 0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0.032064 | 0.045334 | 0.036178 | 0.026792 | 0.053162 | 0.008346 | 0.016285 | |
| 0.008541 | 0.006998 | 0.011037 | 0.006175 | 0.014395 | 0.004243 | 0.00772 | |
| 3.754168 | 6.478501 | 3.27804 | 4.339133 | 3.693014 | 1.966895 | 2.109516 | |
| 0.00027 | 2.14E-09 | 0.001368 | 3E-05 | 0.000335 | 0.051503 | 0.036978 | |
| 0.015154 | 0.031479 | 0.014327 | 0.014567 | 0.02466 | -5.5E-05 | 0.001 | |
| 0.048974 | 0.059188 | 0.05803 | 0.039017 | 0.081664 | 0.016747 | 0.031569 | |
| 120 | 120 | 120 | 120 | 120 | 120 | 120 | |
| 1.97993 | 1.97993 | 1.97993 | 1.97993 | 1.97993 | 1.97993 | 1.97993 | |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | |

1. row names cut: Stata's `c.unemp@1bn.region`
2. column names from `r(table)`: b = rate, se, …
3. Praslin row: lower bound below zero, `-5.5E-05`
4. cell A20: `matrix(r(table))` without transpose or names

# Without names and without the transpose

```
. putexcel A20 = matrix(r(table))
```

The same file, cell A20 onward: 9 rows (b, se, t, …) by 7 regions, with no labels. Stata accepted the command; the sheet is unreadable.

1. always copy and transpose: `matrix T = r(table)'`
2. always add `, names`

# Keep only what the release needs

```
. matrix R = T[1..., 1..2], T[1..., 5..6]

. matrix list R

R[7,4]
                    b          se          ll          ul  1
c.unemp@
1.region    .03206382   .00854086   .01515351   .04897412
c.unemp@
2.region    .04533362   .00699755   .03147896   .05918827
c.unemp@
3.region    .03617833   .01103657   .01432668   .05802997
c.unemp@
4.region    .02679227   .00617457   .01456705   .03901748
c.unemp@
5.region    .05316234   .01439538   .02466049   .08166418
c.unemp@
6.region     .0083458   .00424313  -.00005531    .0167469
c.unemp@
7.region    .01628456   .00771957   .00100035   .03156878
```

1 four columns: rate, SE, lower and upper bound

# Percentages and names for the release

```
. matrix rownames P = North Central1 Central2 EastSouth West PraslinLD Perseverance

. matrix list P, format(%5.1f)

P[7,3]
              rate  lower  upper
       North  71.3   66.5   76.1
    Central1  68.7   64.4   72.9
    Central2  58.6   55.3   62.0
   EastSouth  65.3   62.5   68.1
        West  74.0   66.2   81.8
   PraslinLD  68.9   64.0   73.8
Perseverance  67.0   62.5   71.4
```

1. rates multiplied by 100
2. region names as row names

YOUR TURN · TASK 28

# Participation on a second sheet

**DO**

Run as one block:

```
svy, subpop(wap): mean lf, over(region)
matrix P = r(table)'
matrix P = P[1..., 1..2], P[1..., 5..6]
putexcel set "output/estimates_q1.xlsx", ///
    sheet("Participation") modify
putexcel A1 = matrix(P), names
```

**YOU SHOULD SEE**

A second sheet, four columns.

**IF NOT**

`file could not be saved`? Close Excel and run again.

**EXTRA** Add a Notes sheet with the svyset line and the source.

ON YOUR OWN · I

# The participation table in percent

THE QUESTION

Turn the regional participation rates into a matrix with three columns (rate, lower, upper) in percent, with readable region names.

HINT

Copy `r(table)'`, keep columns 1, 5 and 6, multiply by 100, then `matrix colnames` and `matrix rownames`.

ON YOUR OWN · I · ANSWER

```
matrix P = r(table)'
matrix P = 100 * (P[1..., 1], P[1..., 5..6])
matrix colnames P = rate lower upper
matrix rownames P = North Central1 ...
matrix list P, format(%5.1f)
```

```
. matrix rownames P = North Central1 Central2 EastSouth West PraslinLD Perseverance

. matrix list P, format(%5.1f)

P[7,3]
              rate  lower  upper
       North  71.3   66.5   76.1
    Central1  68.7   64.4   72.9
    Central2  58.6   55.3   62.0
   EastSouth  65.3   62.5   68.1
        West  74.0   66.2   81.8
   PraslinLD  68.9   64.0   73.8
Perseverance  67.0   62.5   71.4
```

1. release-ready: percent, one decimal, names

# my_day3.do, top to bottom

```
use "data/lfs_teach.dta", clear

svyset psu [pweight = wgt_final], strata(district)

generate wap = (ilo_age >= 15)

generate lf = (ilo_lfs19 == 1 | ilo_lfs19 == 2)

generate unemp = (ilo_lfs19 == 2)

svy: ratio unemp/lf

svy, subpop(lf): mean unemp, over(region)

matrix T = r(table)'

estat cv

putexcel set "output/estimates_q1.xlsx", sheet("Unemployment") replace

putexcel A1 = matrix(T), names
```

# These are NBS's Q1 2026 labour estimates, produced and documented by one do-file.

QUICK CHECK

## You run the estimate, then `estat cv`, then `matrix T = r(table)'`. What is in T?

A. the rates

B. the CV results, or an error

C. nothing

ANSWER

## You run the estimate, then `estat cv`, then `matrix T = r(table)'`. What is in T?

A the rates

**B** the CV results, or an error

C nothing

Every command that stores results replaces the last ones. Copy `r(table)` first.

QUICK CHECK

## In `r(table)`, which rows hold the 95% interval?

A. b and se

B. ll and ul

C. t and pvalue

ANSWER

## In `r(table)`, which rows hold the 95% interval?

A. b and se

**B. ll and ul**

C. t and pvalue

ll = lower limit, ul = upper limit (log M01).

# Part 8 on one page

**CHEAT CARD · KEEP THIS ONE**

| | |
|---|---|
| `matrix list r(table)` | see the last estimate's results |
| `matrix T = r(table)'` | copy, regions as rows |
| `matrix R = T[1..., 1..2], T[1..., 5..6]` | keep columns |
| `matrix P = 100 * P` | percent |
| `matrix rownames P = a b c` | names for the release |
| `putexcel A1 = matrix(T), names` | into the open sheet |

# Practice on your own

**PRACTICE SET · practice_D3.do, PART 8**

CORE P1 List `r(table)` after the regional rates: how many rows and columns?

CORE P2 Copy and transpose it; type the SE of West, four decimals.

CORE P3 Write it to `output/p8_estimates.xlsx`.

EXTENSION P4 Keep rate, SE, lower and upper only.

CHALLENGE P5 Participation in percent with region names; type North's upper bound.

Each exercise has a CHECK line: silent means correct. Solutions are released at the end of the day.

# Today’s product is done: official estimates with their margins of error.

lost? open `catchup_D3_S8.do`

# Lab 2 · The estimates release

**THE SITUATION**

The labour statistics section prepares the Q1 2026 release: unemployment and participation by sex and age group, with intervals and CVs, in one Excel file.

**OPEN**

`lab_D3_2.do` in `C:\NBS_Stata\day3`. Close Excel before each run.

**THE TASKS**

**CORE** Tasks 1 to 5: design, flags, rates by sex and age group, CVs, Excel

**EXTENSION** Task 6: flag rates above NBS's CV threshold

**CHALLENGE** Task 7: time-related underemployment rate

**CORRECT WHEN**

the file runs without red text and `output/release_q1.xlsx` has three sheets.

# The lab file: lab_D3_2.do

```
* TASK 3 (Core)  Unemployment by sex and by age group
*   Estimate the unemployment rate by sex and copy r(table)' into S;
*   then by age group into A. Run estat cv after the age groups.
* >>> YOUR CODE



* CHECK Task 3 (silent = correct)
assert abs(S[2, 1] - .0317034) < 1e-6
assert abs(A[1, 1] - .1277627) < 1e-6
```

Close Excel before each run: Tasks 5 and 6 write to `output/release_q1.xlsx`.

# Green note when your last check is silent. Then the Challenge task.

# Day 3, done

**YOU CAN NOW**

- check NBS's weights against the design
- declare the design with `svyset`
- estimate means, totals, shares and rates
- estimate for groups with `subpop()` and `over()`
- judge precision with CV and DEFF
- write the estimates table to Excel

**TODAY'S PRODUCT**

`estimates_q1.xlsx`

**TOMORROW**

Who is more likely to be unemployed: models with `svy:`. Prices over time.

# Daily check

1. Open `check_D3.do` from the day3 folder.
2. Type your **participant code** on the first line.
3. One sentence: what was unclear today?
4. Close Excel. Save and run with **Ctrl + D**.

Read the last line: how many of the checks are done. It is not a grade.

# Tomorrow we ask why: which groups face higher unemployment.